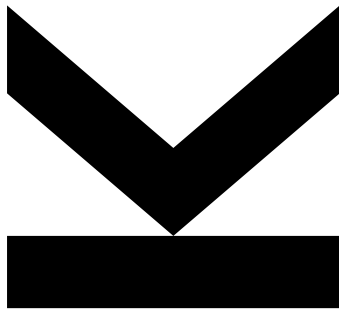


Conditional Diffusion Models as Generative Classifiers for Out-of-Distribution Detection in Inkjet Print Quality Control



Master Thesis

to obtain the academic degree of

Master of Science

in the Master's Program

Artificial Intelligence

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Submission
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May 2026

Abstract

Out-of-distribution (OOD) detection is a critical challenge for safe machine learning. Deep models often face data outside their training distribution while still producing overconfident predictions. This thesis treats conditional diffusion models as generative classifiers: class-conditional reconstruction error serves as the OOD signal.

Unlike likelihood-based approaches that can be brittle in high-dimensional image spaces, diffusion models ask how well a sample can be denoised under competing class conditions. We develop a binary conditional diffusion model (CDM) for CIFAR-10 OOD detection on a within-CIFAR *airplane-vs-rest* split (single ID class; multi-class generalisation is left for future work) and introduce a class-conditional separation loss that maximises the distance between class-conditional noise predictions, indirectly widening reconstruction-error gaps. The central result is that the separation loss improves both performance and stability. Without the separation loss ($\lambda = 0$, seeds 42, 123, and 456), the binary CDM reaches $92.52\% \pm 11.07\%$ AUROC, revealing strong seed sensitivity; with $\lambda = 0.02$ over the same three seeds, performance rises to $99.03\% \pm 0.07\%$, a 6.5 percentage-point gain with far lower variance. These results are fully reproducible from released public artefacts: seed-42 achieves 98.98% AUROC on the within-CIFAR airplane-vs-rest split and 90.50–96.97% AUROC across five external datasets (CIFAR-100, Places365, FashionMNIST, SVHN, Textures).

We further test whether the same principle transfers to industrial quality classification of inkjet print samples. We introduce a multi-head conditioning mechanism for structured manufacturing data and evaluate the public Inkjet00D pipeline on the public FTI_Zer0P dataset. Under 5-fold cross-validation, the crop-based CDM reaches 0.8673 ± 0.0230 AUROC at $\lambda = 0$. In contrast to CIFAR-10, non-zero separation-loss weights do not significantly improve performance, revealing a boundary condition: separation helps when class-conditional noise-prediction gaps translate into reconstruction-error gaps, but it does not automatically transfer to small, fine-grained industrial datasets. Per-feature analysis shows strong heterogeneity: distance and dot features are easiest, while edge roughness features remain harder and more variable.

We analyse why diffusion-based reconstruction error succeeds where other generative approaches often fail, characterise the Monte Carlo timestep trade-off ($K = 10$ provides a strong balance at about $68.6\times$ the ResNet-18 wall-clock cost on an NVIDIA GeForce RTX 2080 Ti; the K -AUROC curve was benchmarked at $\lambda = 0.01$, and transferability to $\lambda = 0.02$ was not independently verified), and provide modular public implementations using PyTorch Lightning and Hydra. For CIFAR-10 OOD detection, training objective design, specifically the separation loss, is the dominant factor; the inkjet study shows that this benefit does not automatically transfer and must be validated rather than assumed.

Keywords: Diffusion Models, Out-of-Distribution Detection, Generative Classification, Industrial Quality Control, Inkjet Printing

Acknowledgements

I would like to express my sincere gratitude to my thesis supervisor, Prof. Dr Sepp Hochreiter, for his valuable guidance, insightful feedback, and unwavering support throughout this research. His expertise in machine learning and deep learning has been instrumental in shaping this work.

I am grateful to my assistant supervisor, Claus Hofmann, MSc, for his constructive feedback and for our back-and-forth discussions, which helped refine the methodology and experimental approach.

This thesis was conducted as an industrial research project at PROFACTOR GmbH, and I am especially grateful to Dr Christian Eitzinger, machine vision team leader at PROFACTOR, who patiently guided me throughout the entire journey. I would also like to thank Abdalla Shahin, Boris Buchroithner, and all colleagues at PROFACTOR for their support and contributions throughout this work.

I would also like to thank the Institute for Machine Learning at Johannes Kepler University Linz for providing the computational resources and supportive research environment necessary to conduct this work.

The work of this thesis was carried out in the premises of PROFACTOR GmbH, where I had the opportunity to access all the needed resources and facilities. This thesis was supported by the Government of Upper Austria under grant Zer0P.

I am deeply indebted to my family and my wife for their relentless support.

Finally, I would like to express my appreciation to my friends for their constant inspiration and encouragement throughout the research and writing process.

During the preparation of this thesis, AI-based writing tools were used for language editing and proofreading. All scientific claims, experimental decisions, source usage, and final wording were reviewed by the author, who remains fully responsible for the originality, correctness, citation practice, and final content of this thesis.

Declaration

I hereby declare that this thesis is my own work and that I have not submitted it, either in its entirety or in substantial part, to any other university for a degree. All sources and external materials used in the preparation of this thesis have been acknowledged and properly cited.

Linz, May 2026

Ahmed Mohammed

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1. Introduction

1.1. Motivation and Problem Statement

Deep learning has achieved considerable success across a wide range of tasks, from image classification and object detection to natural language processing and autonomous decision-making. However, most deployed models assume that test data is drawn from the training distribution, an assumption frequently violated in practice. Models encounter inputs that differ fundamentally from their training distribution: novel object categories, sensor degradation, adversarial perturbations, or simply data from a different domain. When this occurs, standard neural networks tend to produce overconfident predictions rather than signalling uncertainty, creating a critical failure mode for safety-critical applications (Hendrycks and Gimpel, 2017; Guo et al., 2017).

Out-of-distribution (OOD) detection addresses this challenge by equipping models with the ability to distinguish in-distribution (ID) data from data that does not belong to the learnt distribution. A reliable OOD detector enables a model to say “I don’t know” when confronted with unfamiliar inputs, a prerequisite for deploying machine learning systems in healthcare, autonomous driving, industrial quality control, and other high-stakes domains (J. Yang et al., 2024).

1.1.1. Limitations of Existing Approaches

Existing OOD detection methods can be broadly categorised into discriminative and generative approaches, each with significant limitations.

Discriminative approaches repurpose features or outputs of classifiers trained for the in-distribution task. Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2017) uses the classifier’s confidence as an OOD score, but it suffers from the well-documented overconfidence problem. ODIN (Liang et al., 2018) applies temperature scaling and input perturbations to improve separation. The Mahalanobis distance method (Lee et al., 2018) uses feature-space distances under the assumption that the data follow a Gaussian distribution. Energy-based scoring (W. Liu et al., 2020) interprets classifier logits as energy values, providing a theoretically motivated OOD signal. While

computationally efficient, these methods rely on classifier-derived representations that may not capture all distributional structure relevant for OOD detection.

Generative approaches aim to model the data distribution directly, which, in principle, should provide a natural OOD signal: samples with low likelihood under the learnt distribution should be flagged as OOD. However, this intuition fails in practice due to the likelihood paradox. Nalisnick et al. (2019) demonstrated that deep generative models, including variational autoencoders (VAEs) and normalising flows, can assign higher likelihoods to OOD data than to their own training distribution. For example, a model trained on CIFAR-10 assigns a higher likelihood to Street View House Numbers (SVHN) images than to CIFAR-10 images. This paradox, linked to the interaction between high-dimensional geometry and model capacity (Kirichenko et al., 2020), undermines the reliability of likelihood-based OOD detection.

Reconstruction-based methods, which measure the deviation between the input and the reconstruction, offer an alternative. VAE-based anomaly detectors use reconstruction probability as a score (An and Cho, 2015), but their pixel-space reconstructions are often too blurry, owing to the Gaussian decoder assumption, to discriminate subtle distributional differences, limiting their effectiveness on complex natural-image benchmarks.

1.1.2. The Promise of Diffusion Models

Diffusion models (Ho, Jain, et al., 2020; Y. Song et al., 2021) have emerged as a powerful class of generative models that learn to reverse a gradual noising process. Starting from pure Gaussian noise, they iteratively denoise samples to produce high-fidelity outputs, achieving strong benchmark results in image generation (Dhariwal and A. Nichol, 2021; Rombach et al., 2022). Crucially, their training objective, predicting the noise added at each timestep, provides a natural reconstruction error signal that is fundamentally different from VAE reconstruction.

The Diffusion Classifier framework introduced by A. C. Li et al. (2023) demonstrates that conditional diffusion models can perform classification without explicit discriminative training. By comparing noise prediction errors under different class-conditional models, the framework selects the class that best explains the observed data. This principle extends naturally to OOD detection: in-distribution samples should yield lower reconstruction error than OOD samples, as the diffusion model has learnt to denoise only data from the training distribution.

This thesis studies a binary proxy-OOD formulation rather than the standard all-class OOD protocol: CIFAR-10 airplane is treated as the ID class, the remaining CIFAR-10 classes form a labelled non-airplane proxy class during training, and external datasets are used for additional OOD evaluation. This framing is closer to single-class novelty detection with an explicit proxy class than to fully unsupervised OOD detection without OOD labels.

This reconstruction-based approach sidesteps the specific failure mode of raw likelihood in high dimensions, though reconstruction error may have its own failure modes under adversarial inputs. Rather than estimating density in high-dimensional space, where pathological behaviours emerge, diffusion models measure how well they can denoise an input given learnt class-conditional data manifolds. The iterative denoising process provides implicit regularisation that makes reconstruction error more resilient to the high-dimensional structure that undermines likelihood-based methods (Wyatt et al., 2022; Graham et al., 2023).

1.1.3. Research Gap

Despite the theoretical appeal, several critical questions remain unexplored:

- How effective are binary conditional diffusion models as OOD detectors across diverse near- and far-OOD benchmark datasets?
- How does the training objective shape OOD detection performance? Specifically, can enforcing class-conditional separability in reconstruction error, through a dedicated separation loss, significantly improve detection reliability?
- What is the best-tested number of Monte Carlo timestep trials K , and how does K govern the accuracy-efficiency frontier?
- What are the practical trade-offs in computational cost, inference speed, and deployment complexity?
- Can the same principles extend beyond standard benchmarks to real-world industrial applications, such as detecting manufacturing quality defects?

This thesis addresses these questions through a systematic investigation of conditional diffusion models for OOD detection, combining rigorous benchmarking on standard datasets with exploratory application to industrial quality control.

1.2. Research Questions and Objectives

1.2.1. Primary Research Question

How can conditional diffusion models be effectively leveraged as generative classifiers for out-of-distribution detection, and what design choices, particularly regarding the training objective and inference strategy, most strongly influence detection performance?

This overarching question is decomposed into the following sub-questions:

1. **RQ1: Effectiveness.** Can reconstruction error from conditional diffusion models serve as a reliable and high-quality OOD detection signal?

2. **RQ2: Separation Loss.** How does incorporating a class-conditional separation loss (a training objective that maximises the distance between ID-conditioned and OOD-proxy-conditioned noise predictions, formalised in Chapter 3) into the diffusion training objective affect OOD detection performance? Can this objective indirectly widen reconstruction-error gaps between ID and non-ID classes and improve detection?
3. **RQ3: Monte Carlo Trials.** What is the relationship between the number of Monte Carlo timestep trials K and the accuracy-efficiency trade-off? At what value of K does performance plateau?
4. **RQ4: Scoring Strategy.** How does the choice of OOD scoring strategy, using ID reconstruction error alone versus contrastive difference scoring (subtracting ID error from OOD error), affect detection performance across near- and far-OOD datasets?
5. **RQ5: Practical Viability.** What are the computational costs of diffusion-based OOD detection relative to discriminative baselines, and under what conditions is the accuracy-cost trade-off favourable?
6. **RQ6: Industrial Application.** How effective are conditional diffusion models for publicly reproducible industrial quality classification, and how do separation loss, feature type, and cross-validation variability affect performance in practical manufacturing settings?

1.2.2. Specific Objectives

To address the research questions, this thesis pursues four objectives:

Objective 1: Framework Development. Design and implement a framework for diffusion-based OOD detection, including:

- A binary conditional diffusion model with class-conditional reconstruction error scoring
- A separation loss training objective that encourages distinct reconstruction error distributions for ID and non-ID classes
- Configurable OOD scoring via Monte Carlo timestep sampling with adjustable trial count K and contrastive difference scoring

Objective 2: Benchmarking. Conduct an evaluation on standard OOD benchmarks:

- **In-distribution:** CIFAR-10 airplane class (Krizhevsky, Hinton, et al., 2009)
- **Within-CIFAR OOD:** remaining nine CIFAR-10 classes
- **External OOD (core reproducible set):** CIFAR-100 (near), Places365 (far), Fashion-MNIST (far), SVHN (far), Textures (far)

- Multi-seed reporting is used where public repeated runs are available, while single-seed result artefacts are explicitly labelled as such

Objective 3: Systematic Ablation Studies. Validate design choices through controlled experiments:

- Separation loss coefficient ($\lambda \in \{0.0, 0.001, 0.01, 0.02, 0.05, 0.1\}$)
- Number of Monte Carlo timestep trials ($K \in \{1, 5, 10, 25, 50, 100\}$)
- OOD scoring strategy (ID reconstruction error alone vs. contrastive difference scoring)

Objective 4: Reproducible Implementation. Develop a well-documented, modular codebase:

- Built on PyTorch (Paszke et al., 2019), PyTorch Lightning (Falcon and team, 2019), and Hydra (Yadan, 2019)
- Standardised evaluation pipeline with standard metrics (Area Under the ROC Curve (AUROC), False Positive Rate at 95% True Positive Rate (FPR@95%TPR), Area Under the Precision-Recall Curve (AUPRC))
- Complete experiment tracking and configuration logging

1.3. Contributions

This thesis makes the following contributions:

1. **Systematic evaluation of binary conditional diffusion models for OOD detection.** We demonstrate that a binary conditional diffusion model (CDM) (airplane vs. rest) achieves strong AUROC on the within-CIFAR split and across five external OOD datasets spanning near- and far-OOD scenarios (Chapter 6).
2. **Separation loss as a key training objective.** We introduce a class-conditional separation loss that maximises the distance between class-conditional noise predictions, indirectly widening reconstruction-error gaps and yielding the clearest CIFAR-10 improvement in our study (Chapter 6, Section 6.4).
3. **Inference design choices: MC trials and contrastive scoring.** We characterise the accuracy–efficiency frontier as a function of Monte Carlo trials K , and demonstrate that contrastive difference scoring is essential, since ID-only scoring fails catastrophically on certain external datasets (Chapter 6, Section 6.3).
4. **Application to industrial quality control.** We evaluate a public YOLO+CDM pipeline for inkjet quality classification under 5-fold cross-validation and show that, unlike on CIFAR-10, the separation loss does not produce a statistically significant improvement on this smaller industrial dataset (Chapter 6, Section 6.6).

5. **Reproducible implementation framework.** We provide modular public implementations using PyTorch Lightning and Hydra with standardised evaluation tools. The CIFAR-10/OOD track is released as DiffusionOOD at <https://github.com/ahmed-3m/DiffusionOOD>, with weights at <https://huggingface.co/ahmed-3m/DiffusionOOD>. The inkjet CDM track is released as InkjetOOD at <https://github.com/ahmed-3m/InkjetOOD> (Mohammed, 2026b), with weights at <https://huggingface.co/ahmed-3m/InkjetOOD>, and uses the public PROFACTOR FTI_Zer0P dataset on Zenodo (Profactor, 2024).

1.4. Thesis Structure

The remainder of this thesis is organised as follows. **Chapter 2** reviews OOD detection methods, diffusion model foundations, industrial quality classification baselines, and positions our contributions via a gap analysis. **Chapter 3** formalises both experimental tracks: the binary CDM with separation loss and contrastive scoring for OOD detection, and the two-stage YOLO+CDM pipeline with multi-head conditioning for inkjet quality control. **Chapter 4** describes the technical realisation, including the software stack, cross-validation infrastructure, and reproducibility measures. **Chapter 5** defines the experimental design, datasets, ablation studies, and evaluation protocols for both tracks. **Chapter 6** presents the experimental results and ablation analyses for the retained public artefacts in the thesis. **Chapter 7** interprets the findings, draws cross-domain insights, and discusses limitations and future work. **Chapter 8** answers the six research questions and summarises contributions.

2. Background and Related Work

This chapter provides the theoretical and empirical context for the two experimental tracks developed in this thesis. We survey the landscape of out-of-distribution (OOD) detection, covering discriminative and generative approaches and their known limitations, then review the diffusion model foundations used throughout. We further describe industrial quality classification baselines and identify the research gaps that motivate our contributions.

2.1. Out-of-Distribution Detection

Out-of-distribution detection addresses a fundamental challenge in deploying machine learning models: identifying when test samples differ significantly from the training distribution. This section formalises the OOD detection problem, characterises different types of distribution shift, and discusses why robust OOD detection is critical for real-world applications.

2.1.1. Problem Definition

Let $\mathcal{X}$ denote the input space and $\mathcal{Y}$ denote the label space. During training, we have access to a dataset $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from an in-distribution (ID) $P_{\text{train}}(x, y)$. A model $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ is learnt to minimise some loss function $\mathcal{L}$ over this distribution:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{(x,y) \sim P_{\text{train}}} [\mathcal{L}(f_{\theta}(x), y)] \quad (2.1)$$

At test time, the model may encounter samples x_{test} that come from a different distribution $P_{\text{ood}}(x)$. The goal of OOD detection is to determine whether a given test sample x is drawn from P_{train} or P_{ood} without access to labels from the OOD distribution during training. Formally, we seek a scoring function $s : \mathcal{X} \rightarrow \mathbb{R}$ such that:

$$s(x) \begin{cases} > \tau & \text{if } x \sim P_{\text{ood}} \\ \leq \tau & \text{if } x \sim P_{\text{train}} \end{cases} \quad (2.2)$$

where τ is a threshold that can be calibrated using a held-out validation set from P_{train} .

The effectiveness of an OOD detector is typically evaluated using metrics that assess the separability of ID and OOD scores:

- **Area Under the ROC Curve (AUROC):** Measures the probability that a randomly chosen OOD sample has a higher score than a randomly chosen ID sample. An AUROC of 1.0 indicates perfect separation, while 0.5 represents random guessing.
- **False Positive Rate at 95% True Positive Rate (FPR@95%TPR):** Measures the fraction of ID samples incorrectly classified as OOD when the threshold is set such that 95% of OOD samples are correctly identified. Lower values indicate better performance.
- **Detection Accuracy:** The proportion of correctly classified samples (both ID and OOD) when using an optimal threshold. While simple to interpret, this metric can be misleading when ID and OOD sample sizes are imbalanced.

Scoring convention. A note on sign convention: the OOD literature uses two competing conventions for the direction of scoring functions. In the *ID-positive* convention (common in anomaly detection and binary classification), higher scores indicate higher confidence that a sample is in-distribution; the threshold comparison is $s(x) < \tau$ for OOD rejection (W. Liu et al., 2020). In the *OOD-positive* convention adopted in this thesis, higher scores indicate greater evidence of being out-of-distribution, and $s(x) > \tau$ triggers an OOD flag (Equation 2.2). This follows the convention used in several diffusion-based OOD works (J. Liu et al., 2025; Heng et al., 2024) and is consistent throughout all chapters and reported FPR95 values: the threshold is set so that 95% of OOD samples satisfy $s(x) > \tau$, and FPR95 measures the resulting false positive rate among ID samples.

Distribution shift can take several forms (covariate shift, label shift, or **semantic shift**), each presenting distinct challenges (J. Yang et al., 2024). Our work primarily addresses semantic shift (test classes absent from training) with both **near-OOD** scenarios (e.g., CIFAR-100 when trained on CIFAR-10) and **far-OOD** scenarios (e.g., SVHN, Textures).

2.2. Deep Learning Approaches

The success of deep learning in representation learning has spawned numerous OOD detection methods. These can be broadly categorised into discriminative methods that exploit classifier properties and generative methods that model the data distribution.

2.2.1. Discriminative Methods

Discriminative methods leverage trained classifiers without density modelling. Key baselines include Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2017), ODIN (Liang et al., 2018) (input perturbation with temperature scaling), Mahalanobis distance in feature space (Lee et al., 2018), and Energy-based detection (W. Liu et al., 2020). These methods are computationally efficient but can struggle when OOD samples lie near decision boundaries.

2.2.2. Reconstruction-based and Generative Approaches

Reconstruction-based methods use the discrepancy between input and reconstruction as an OOD indicator. Standard autoencoders measure reconstruction error $\|x - d(e(x))\|^2$ as an anomaly score, but the assumption that autoencoders fail on OOD samples often does not hold: they can generalise to reconstruct OOD inputs well, particularly with high-dimensional latent spaces (Baur et al., 2021). Variational autoencoders (VAEs) suffer from both likelihood misassignment and poor reconstruction quality due to posterior collapse (An and Cho, 2015).

Generative models that explicitly learn $P_{\text{train}}(x)$ should in principle use likelihood for OOD detection, but this intuition fails in practice. Nalisnick et al. (2019) demonstrated the likelihood paradox: deep generative models, including normalising flows with exact likelihoods, can assign higher likelihood to OOD data than ID data (e.g., flows trained on CIFAR-10 assign higher likelihood to SVHN). Kirichenko et al. (2020) showed that likelihood decomposes into on-manifold and off-manifold components, with off-manifold volume often dominating. These findings motivated our investigation of diffusion models, whose reconstruction-based scoring sidesteps the likelihood paradox.

2.3. Diffusion Models

Diffusion models have emerged as a powerful class of generative models, achieving strong benchmark results in image synthesis and other generative tasks (Dhariwal and A. Nichol, 2021; Rombach et al., 2022). This section provides the mathematical foundations necessary for understanding how diffusion models can be applied to OOD detection.

2.3.1. Mathematical Foundations

Diffusion models, also known as denoising diffusion probabilistic models (DDPMs) (Sohl-Dickstein et al., 2015; Ho, Jain, et al., 2020), define a forward noising process and learn to reverse it.

Forward Process

The forward process gradually adds Gaussian noise to data over T timesteps:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}) \quad (2.3)$$

where $\{\beta_t\}_{t=1}^T$ is a variance schedule. Crucially, we can sample x_t directly from x_0 :

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (2.4)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. As $t \rightarrow T$, x_T approaches pure Gaussian noise.

Reverse Process

The reverse process learns to denoise:

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (2.5)$$

The mean μ_θ can be parameterised in various ways. Following DDPM, we typically predict the noise $\epsilon_\theta(x_t, t)$ and compute:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) \quad (2.6)$$

Training Objective

The model is trained by minimising the variational lower bound, which simplifies to:

$$\mathcal{L} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right] \quad (2.7)$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and $t \sim \text{Uniform}(\{1, \dots, T\})$.

This objective has an elegant interpretation: the model learns to predict the noise that was added to clean data x_0 at timestep t .

2.3.2. DDPM Variants and Extensions

Noise Schedules: The variance schedule $\{\beta_t\}$ significantly impacts performance; cosine schedules (A. Q. Nichol and Dhariwal, 2021) maintain more information at higher timesteps than linear schedules.

Score-based Formulation: Y. Song et al. (2021) showed that diffusion models learn the score function $\nabla_x \log p(x)$, revealing the connection $\epsilon_\theta(x_t, t) \approx -\sqrt{1 - \bar{\alpha}_t} \nabla_{x_t} \log p(x_t)$. This perspective connects diffusion models to Langevin dynamics and enables continuous-time stochastic differential equation (SDE) formulations. Accelerated samplers such as Denoising Diffusion Implicit Models (DDIM) (J. Song et al., 2021) allow fewer-step generation without retraining. This thesis uses the discrete DDPM formulation throughout.

2.3.3. Conditional Diffusion Models

Conditioning enables diffusion models to generate samples from specific categories or with desired attributes.

Classifier Guidance

Dhariwal and A. Nichol (2021) introduced classifier guidance, which steers generation using gradients from a separately trained classifier. While effective, it requires maintaining an auxiliary classifier.

Classifier-Free Guidance

Ho and Salimans (2022) proposed classifier-free guidance, which eliminates the need for a separate classifier:

$$\tilde{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t, \emptyset) + w (\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t, \emptyset)) \quad (2.8)$$

During training, the condition y is randomly dropped with probability p (typically 0.1), allowing the model to learn both conditional and unconditional predictions. This approach has become the standard for conditional generation.

Class Embeddings

Conditioning typically injects class information through learnt embeddings added to timestep embeddings or through cross-attention mechanisms in transformer-based architectures.

Multi-Head Conditioning

Beyond single-label conditioning, diffusion models can be conditioned on multiple heterogeneous inputs simultaneously. In our inkjet quality control application, we employ

multi-head conditioning where the model receives template type (categorical), feature type (categorical), quality label (binary), and bounding box coordinates (continuous) as separate conditioning inputs, each encoded through dedicated embedding layers before being combined. This multi-head approach enables a single model to handle diverse feature types and spatial contexts.

Diffusion models have achieved strong results across image generation, super-resolution, inpainting, and text-to-image synthesis (Dhariwal and A. Nichol, 2021; Rombach et al., 2022). However, their application to discriminative tasks like OOD detection and industrial quality classification remains relatively unexplored. A preliminary attempt to apply text-conditioned Stable Diffusion (Rombach et al., 2022) to the inkjet defect dataset yielded unsatisfactory generation quality on this small, domain-specific dataset: the model could not reliably reproduce fine-grained defect patterns from text prompts alone, and the classification stage was never reached. This negative result motivated the shift to class-label conditioning on a lightweight binary CDM trained from scratch, which forms the basis of this work.

2.3.4. Concurrent Diffusion-Based OOD Detection

Several concurrent works explore diffusion models for OOD detection and anomaly scoring, each with a distinct mechanism. Understanding how this thesis differs from these approaches is essential for positioning the separation-loss contribution.

Graham et al. (2023) (Graham et al., 2023) demonstrate that reconstruction error from a pre-trained unconditional diffusion model can serve as an anomaly score for medical images, noising a test input to an intermediate timestep and measuring the residual after denoising. Their method requires no class conditioning and no contrastive scoring; the OOD signal is the absolute reconstruction magnitude rather than a class-conditional difference. This thesis differs in two ways: we use *binary* class-conditional denoising (airplane vs. non-airplane) and a *contrastive* score $e_0 - e_1$ rather than absolute reconstruction error, which our ablation (Section 6.3) shows is essential, since absolute ID error collapses to 20.2% AUROC on SVHN.

DiffGuard (Gao et al., 2023) (Gao et al., 2023) use a pre-trained Stable Diffusion model to measure semantic mismatch between a test image and a text description of the in-distribution class; the OOD score is the CLIP-space residual between the input and its diffusion-guided reconstruction. DiffGuard therefore relies on large-scale pre-training and text conditioning rather than task-specific binary training. In contrast, the binary CDM in this thesis is trained from scratch on a binary classification task and introduces the separation loss to *explicitly* widen the reconstruction gap during training rather than measuring it post-hoc.

Livernoche et al. (2024) (Livernoche et al., 2024) investigate diffusion-based density estimation for anomaly detection, finding that diffusion-model likelihoods (approximated via the ELBO) can be competitive with flow-based likelihoods when properly

normalised. Their study is complementary to this work: they focus on likelihood estimation, whereas we focus on reconstruction-error contrast and demonstrate that the training objective (separation loss) dominates the scoring signal. Their negative findings on raw likelihood further motivate the reconstruction-error contrastive approach taken here.

Wu et al. (2024) (Wu et al., 2024) show that a diffusion model’s noise-prediction network is implicitly a noise classifier and that adding a contrastive training objective in noise-prediction space improves both generation and OOD discrimination. This is the closest prior work to the separation loss proposed in Chapter 3: both operate on noise-prediction differences under competing conditions. The key differences are scope and evaluation: Wu et al. study the contrastive objective as a training regulariser for generation quality, while this thesis isolates it as the primary OOD detection lever via a systematic λ ablation, and further tests its transfer to an industrial domain.

Table 2.1 summarises the key design choices across these works and this thesis.

Table 2.1.: Comparison of concurrent diffusion-based OOD detection methods. CFG = classifier-free guidance; $\checkmark$ = feature present; $\times$ = feature absent. “Benchmark protocol” refers to the headline evaluation setup; “OOD type” indicates whether the method was primarily evaluated on near-OOD (semantic shift), far-OOD (low-level statistic shift), or domain-specific anomaly inputs.

Method	Training	Scoring	Class cond.	Sep. loss	Benchmark protocol	OOD type
Graham et al. 2023 (Graham et al., 2023)	Pre-trained	Abs. reconstruction	$\times$	$\times$	Medical (BraTS, CXR)	Domain-specific
DiffGuard 2023 (Gao et al., 2023)	Pre-trained	CLIP residual	Text (CFG)	$\times$	ImageNet near/far-OOD	Near + far
Livernoche et al. 2024 (Livernoche et al., 2024)	From scratch	Likelihood (ELBO)	$\times$	$\times$	CIFAR-10 vs. SVHN, CelebA	Far-OOD
Wu et al. 2024 (Wu et al., 2024)	From scratch	Contrastive noise	$\checkmark$	$\checkmark$	CIFAR-10/100 multi-class	Near + far
This work	From scratch	Contrastive diff. score	Binary	$\checkmark$	CIFAR-10 airplane-vs-rest + 5 ext. sets	Near + far

2.4. Low-Dimensional Data Geometry

The low-dimensional concentration view posits that high-dimensional data concentrate on or near lower-dimensional structures embedded in the ambient space ($d \ll D$; for CIFAR-10, $D = 3072$ while intrinsic dimension is typically in the tens to low hundreds (Pope et al., 2021)). This view has direct implications for OOD detection.

Different generative models capture manifold structure differently: VAEs parameterise it via a latent space, flows compute exact likelihood but conflate manifold density with

ambient volume (Kirichenko et al., 2020; Le Lan and Dinh, 2021) (explaining the likelihood paradox), and diffusion models iteratively project noisy samples onto the learnt manifold through progressive denoising (Karras et al., 2024).

The manifold perspective explains why likelihood-based OOD detection fails (volume dominates distance in high dimensions (Fefferman et al., 2016; Pope et al., 2021)) and suggests reconstruction error as an alternative: the distance from a sample to the learnt manifold provides a more reliable OOD signal than density. Diffusion models' iterative denoising can be interpreted as progressive projection onto the data manifold, where high reconstruction error indicates that a sample lies far from the learnt manifold (Wyatt et al., 2022). This perspective motivates our use of reconstruction error from conditional diffusion models as an OOD detection signal.

2.5. Industrial Quality Control and Anomaly Detection

As a practical application domain for the principles developed in this thesis, we investigate automated quality classification for inkjet print samples. This section reviews the relevant background on industrial visual inspection, object detection for region-of-interest extraction, and anomaly detection methods used as baselines.

2.5.1. Automated Visual Inspection

Automated visual inspection (AVI) has become essential in modern manufacturing, replacing subjective human assessment with consistent, quantitative evaluation (Czimmermann et al., 2020). In inkjet print samples, quality defects manifest as misaligned features, irregular edge roughness, incorrect spacing, and malformed dot patterns. These defects reduce production yield and compromise the feedback loop used to optimise printing parameters.

Traditional AVI systems rely on hand-crafted features and template matching, which require significant domain expertise and are brittle to variations in printing conditions. Deep learning-based approaches have largely superseded these methods, leveraging learnt representations to capture subtle quality variations that resist manual feature engineering (Akçay et al., 2022).

A key challenge in industrial quality classification is choosing the input representation. Full-image approaches preserve global spatial context (the position of features relative to page edges and other features), while crop-based approaches focus local analysis on individual regions of interest. The retained public inkjet experiments in this thesis evaluate the crop-based YOLO+CDM path; broader full-image versus crop comparisons are left outside the retained evidence base unless released with public code, splits, and results.

2.5.2. Object Detection for Feature Extraction

Modern object detection architectures, particularly the YOLO (You Only Look Once) family (Redmon et al., 2016), enable real-time detection and localisation of objects in images. In our pipeline, we use YOLOv8 (Jocher et al., 2023) to detect and extract individual printed features from full template images before quality classification.

YOLOv8 Architecture

YOLOv8 uses a CSPDarknet backbone with a Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) for multi-scale feature fusion. The anchor-free detection head simultaneously predicts bounding box coordinates and class probabilities. The YOLOv8n (nano) variant used in our work contains approximately 3.2M parameters and achieves real-time inference.

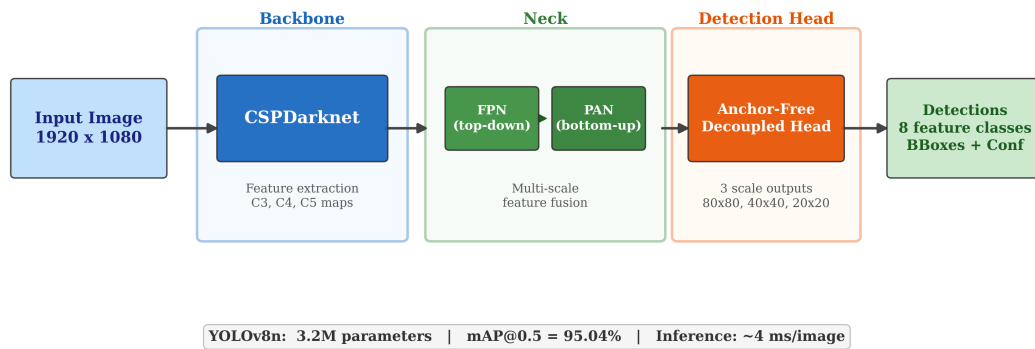


Figure 2.1.: YOLOv8 architecture overview. The CSPDarknet backbone extracts multi-scale feature maps (C3, C4, C5), which are fused by the Feature Pyramid Network (top-down) and Path Aggregation Network (bottom-up). The anchor-free decoupled detection head produces bounding boxes, class predictions, and confidence scores at three scales. The YOLOv8n variant (3.2M parameters) achieves mAP@0.5 of 95.04% on our inkjet feature detection task.

Role in the Pipeline

Object detection serves as a preprocessing stage that transforms full template images (1920×1080 pixels) into individual feature crops (128×128 pixels). In the retained CDM evaluation, the processed source-image groups are expanded into 6,408 feature-level samples. Each detected region carries metadata including template type (A, B, or C), feature class (8 types), and bounding box coordinates, which serve as conditioning inputs for the diffusion model.

2.5.3. Anomaly Detection Methods

Anomaly detection methods learn only from normal (defect-free) samples and detect anomalies as deviations from the learnt normality. They provide relevant context for industrial visual inspection, although non-CDM baseline experiments are not retained as thesis evidence unless they are reproducible from public artefacts.

- **PatchCore** (Roth et al., 2022): A memory-bank approach that scores anomalies by computing the minimum distance from test patch features to a coreset of pre-trained ResNet features from normal samples. Requires no training beyond feature extraction.
- **PaDiM** (Defard et al., 2021): Models normal patch features as position-wise multi-variate Gaussians; anomaly scores use Mahalanobis distance from the learnt distribution.
- **Student-Teacher Feature Pyramid Matching (STFPM)** (G. Wang et al., 2021): Knowledge distillation where a student mimics a pre-trained teacher on normal data; anomalies produce feature-pyramid discrepancies.
- **Autoencoder**: Reconstruction-based approach where reconstruction error serves as the anomaly score, an assumption that can be unreliable for subtle quality defects.

Recent extensions include FR-PatchCore (Jiang et al., 2024) for spatial robustness; J. Wang et al. (2025) survey the growing application of diffusion models to industrial anomaly detection; X. Li et al. (2025) contribute a continual diffusion method that handles multiple anomaly types within a single model.

Diffusion-Based Anomaly Detection

A parallel line of work applies DDPM-style generative models directly to industrial anomaly detection, and is the closest antecedent to the inkjet pipeline developed in Chapter 3. AnoDDPM (Wyatt et al., 2022) introduced partial-diffusion anomaly scoring, noising a test image only up to an intermediate timestep and using the reconstruction residual between the original and the denoised output as the anomaly map; it demonstrated that unconditional diffusion reconstructions are competitive with feature-distance baselines on MVTec-AD without requiring any defect labels. In contrast, EfficientAD (Batzner et al., 2024) takes a discriminative student-teacher route with an explicit efficiency constraint, achieving millisecond-level inference and state-of-the-art MVTec-AD numbers while using no diffusion component; it therefore serves as a strong non-diffusion reference point for the cost-accuracy trade-off discussed in Chapter 7. Finally, Heckler-Kram et al. (2026) release MVTec-AD 2, a substantially harder successor benchmark with real (non-synthetic) defects and multiple lighting conditions; they show that several top methods on the original MVTec-AD lose a large fraction of their AUROC when moved to MVTec-AD 2, highlighting that headline anomaly-detection

numbers are benchmark-specific. Our binary CDM differs from all three: it is explicitly conditioned on a binary class variable and trained with a separation loss (Equation 3.23 in Chapter 3) rather than a purely unconditional reconstruction objective, and it produces a contrastive score across two class conditions rather than an unconditional residual.

2.5.4. Supervised Classification Methods

When labelled data for both good and defective samples is available, supervised classification methods provide a strong alternative. Common architectures considered in this application area include:

- **ResNet-FullImg:** A standard ResNet-18 (He et al., 2016) pre-trained on ImageNet, fine-tuned on full template images resized to 224×224 . This approach preserves global spatial context, allowing the model to leverage inter-feature relationships and page-level alignment cues.
- **ResNet-CropOnly:** The same ResNet-18 architecture applied to YOLO-cropped feature regions resized to 224×224 . This isolates local feature quality but loses global context.
- **Dual-Branch:** Two ResNet-18 encoders process full images and cropped regions in parallel, with feature embeddings concatenated and passed through a fusion multi-layer perceptron (MLP) for final classification. A feature-type embedding provides additional conditioning.

These supervised methods serve as upper bounds for the quality classification task, establishing the achievable performance with direct label supervision.

2.6. Gap Analysis and Positioning

We now identify gaps in the current literature and position our contributions.

Limitations and gaps. Existing likelihood-based generative OOD detectors suffer from the likelihood paradox; many widely used discriminative detectors (MSP, ODIN, Mahalanobis, Energy) train on ID data only but rely on temperature, perturbation, or threshold settings that are typically calibrated against held-out OOD samples and remain miscalibrated under stronger distributional shift. Concurrent diffusion-based OOD approaches (Section 2.3.4) address the likelihood paradox via reconstruction-error scoring or pre-trained guidance, but lack systematic design studies, particularly on how training objectives and inference strategies affect AUROC in controlled settings. Four gaps remain open: (1) no principled study of separation-loss objectives for binary conditional

diffusion OOD detection (Wu et al. (Wu et al., 2024) study contrastive training for generation quality but do not ablate the separation signal as the primary OOD lever); (2) contrastive scoring strategies are underexplored; (3) Monte Carlo trial budgets are unstudied; and (4) transfer conditions from large benchmarks to small industrial datasets are unknown.

Thesis positioning. **Track 1** addresses Gaps 1–3 via a binary CDM separation-loss ablation on CIFAR-10, the K-sweep, and a scoring-method comparison, all with fully public code and artefacts (Mohammed, 2026a). **Track 2** addresses Gap 4: a public YOLO+CDM pipeline from InkjetOOD (Mohammed, 2026b), evaluated on the FTI_Zer0P dataset under 5-fold cross-validation, identifies the boundary conditions under which the separation-loss gain fails to transfer.

Practical Contribution: We provide modular implementations built on PyTorch Lightning (Falcon and team, 2019) and Hydra (Yadan, 2019) with evaluation tools, reproducibility measures, and documentation. The CIFAR-10/OOD benchmark code is publicly available at <https://github.com/ahmed-3m/DiffusionOOD>, with weights at <https://huggingface.co/ahmed-3m/DiffusionOOD>. The inkjet CDM implementation is publicly available at <https://github.com/ahmed-3m/InkjetOOD> (Mohammed, 2026b), with weights at <https://huggingface.co/ahmed-3m/InkjetOOD>, and uses the PRO-FACTOR FTI_Zer0P dataset released on Zenodo under CC BY 4.0 (Profactor, 2024).

Our work explores whether diffusion models’ generative capabilities translate into discriminative signals for OOD detection and quality classification, contributing systematic ablations and practical cross-domain insights to this rapidly evolving field.

3. Methodology

This chapter develops the methodological framework for both experimental tracks. We formalise the out-of-distribution detection problem as a binary generative classification task, introduce the class-conditional separation loss that is the primary novel contribution of this work, and describe the two-stage inkjet quality control pipeline with its multi-head conditional diffusion model. Formal analysis and architectural details are provided throughout to ground the experimental choices made in Chapters 4 and 6.

3.1. Problem Formulation

This section formalises the OOD detection problem and establishes how conditional diffusion models can be used as generative classifiers for this task.

3.1.1. Relation to One-Class Novelty Detection

Our problem formulation differs from the standard multi-class OOD detection benchmark setting. In the standard protocol (e.g., OpenOOD (Zhang et al., 2023)), a multi-class classifier is trained on all classes of an in-distribution dataset (e.g., all 10 CIFAR-10 classes), and OOD detection is evaluated on samples from entirely different datasets. By contrast, our binary CDM models a single target class (airplane) against all others, which is closer to the one-class novelty detection (also called one-class classification or anomaly detection) paradigm (Ruff et al., 2018; Tack et al., 2020).

In one-class novelty detection, a model learns the distribution of a single “normal” class and flags deviations as anomalous. Classical approaches include One-Class SVM (Schölkopf et al., 2001), Deep SVDD (Ruff et al., 2018), and contrastive methods such as CSI (Tack et al., 2020) and PANDA (Reiss, Cohen, et al., 2021). Our binary CDM extends this paradigm by explicitly modelling both the target class and an OOD proxy class, enabling contrastive scoring rather than relying on a single normality model. This dual-condition design is the key architectural distinction from pure one-class methods and is what enables the separation loss to amplify the reconstruction gap.

Consequently, direct numerical comparison with standard multi-class OOD benchmarks (MSP, ODIN, Energy on all-class CIFAR-10) would be misleading, as they solve a differ-

ent problem. Instead, we contextualise our results against one-class baselines evaluated on the same per-class CIFAR-10 protocol in Section 6.2.

3.1.2. Mathematical Definition

With the general OOD detection formulation established in Chapter 2, we now focus on our approach using conditional diffusion models.

Binary Training Setup: We assume access to a labelled training dataset $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$ where $x_i \in \mathbb{R}^{3 \times 32 \times 32}$ are CIFAR-10 images and $y_i \in \{0, 1\}$ are binary labels: $y = 0$ for the in-distribution class (airplane, CIFAR-10 class 0) and $y = 1$ for the OOD-proxy class (all other nine CIFAR-10 classes pooled together). The data is drawn i.i.d. from the training distribution $P_{\text{train}}(x, y)$. Consistent with the OOD-positive scoring convention stated in Section 2.1.1, higher values of the scoring function $s_{\text{OOD}}(x)$ indicate greater evidence of being out-of-distribution.

Test Setup: At test time, we receive samples x_{test} from either:

- **In-distribution (ID):** $x_{\text{test}} \sim p(x \mid y = 0)$, i.e. airplane images.
- **OOD:** $x_{\text{test}} \sim P_{\text{ood}}(x)$, where P_{ood} is either the within-CIFAR OOD proxy ($y = 1$ pool) or an unseen external dataset. In the external benchmark protocol, these external samples are evaluated against the full CIFAR-10 test reference pool because the binary CDM was trained on both conditional branches of CIFAR-10 (airplane as $c=0$, non-airplane as $c=1$). The core reproducible external set comprises CIFAR-100, Places365, FashionMNIST, SVHN, and Textures.

Binary Generative Classification Framework: We train a single conditional diffusion model to learn the two class-conditional distributions $p_{\theta}(x \mid c)$ for $c \in \{0, 1\}$. (Here and throughout, c denotes the binary class condition, generalising the training label y : $c = 0$ refers to the ID class and $c = 1$ to the OOD-proxy class.) For OOD detection we exploit the observation that ID samples should be reconstructed accurately under the ID condition ($c = 0$) but poorly under the OOD condition ($c = 1$), and vice versa for OOD samples.

Per-Condition Reconstruction Errors: For a test sample x we compute reconstruction errors under each condition:

$$e_0(x) = \mathbb{E}_{t \sim \mathcal{U}[1, T], \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c=0 \right) \right\|^2 \right] \quad (3.1)$$

$$e_1(x) = \mathbb{E}_{t \sim \mathcal{U}[1, T], \epsilon \sim \mathcal{N}(0, I)} \left[\left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c=1 \right) \right\|^2 \right] \quad (3.2)$$

where $\epsilon_{\theta}(x_t, t, c)$ is the noise-prediction UNet conditioned on binary class c .

OOD Scoring Function (Difference): The primary scoring method computes the signed difference between ID-conditioned and OOD-conditioned errors:

$$s_{\text{OOD}}(x) = e_0(x) - e_1(x) \quad (3.3)$$

A high score indicates the model denoises x far better under the OOD condition than the ID condition, which is strong evidence that x is out-of-distribution. Samples with $s_{\text{OOD}}(x) > \tau$ (for threshold τ) are classified as OOD.

Alternative Scoring Methods: We also investigate:

$$s_{\text{ratio}}(x) = \frac{e_0(x)}{e_1(x)} \quad (3.4)$$

$$s_{\text{id-only}}(x) = e_0(x) \quad (3.5)$$

The ratio formulation normalises by the magnitude of the OOD-conditioned error, making it more robust to scale differences across OOD distributions. The ID-only formulation ignores the OOD-conditioned error entirely; we include it as an ablation to measure the contribution of contrastive scoring. Section 6.3 shows that comparing the two conditional errors (difference or ratio) is essential: ID-only scoring collapses to 20.2% AUROC on SVHN.

3.1.3. OOD Scoring Algorithm

Algorithm 1 details the procedure for computing OOD scores at inference time.

Algorithm 1 *Binary CDM OOD Scoring (Difference Method).*

Require: Test image $x \in \mathbb{R}^{3 \times 32 \times 32}$, trained model ϵ_θ

Require: Number of MC trials K , timesteps T , threshold τ

Ensure: OOD score $s(x)$, decision $\text{is_ood} \in \{0, 1\}$

1: $\hat{e}_0 \leftarrow 0; \hat{e}_1 \leftarrow 0$

2: **for** trial $k = 1$ to K **do**

3: $t_k \sim \text{Uniform}\{1, \dots, T\}; \epsilon_k \sim \mathcal{N}(0, I)$

4: $x_{t_k} \leftarrow \sqrt{\bar{\alpha}_{t_k}} x + \sqrt{1 - \bar{\alpha}_{t_k}} \epsilon_k$

5: $\hat{e}_0 \leftarrow \hat{e}_0 + \|\epsilon_k - \epsilon_\theta(x_{t_k}, t_k, c=0)\|^2$

6: $\hat{e}_1 \leftarrow \hat{e}_1 + \|\epsilon_k - \epsilon_\theta(x_{t_k}, t_k, c=1)\|^2$

7: **end for**

8: $\hat{e}_0 \leftarrow \hat{e}_0 / K; \hat{e}_1 \leftarrow \hat{e}_1 / K$

9: $s(x) \leftarrow \hat{e}_0 - \hat{e}_1$ (positive: OOD; negative: ID)

10: **if** $s(x) > \tau$ **then return** OOD **else return** ID

Complexity Analysis: The cost of Algorithm 1 is $O(2K \cdot N)$ where N is the cost of one forward pass. For $K = 50$ and the binary UNet ($\approx 68.8\text{M}$ parameters), the measured timing is 4861.1 s (about 81.0 minutes) per 10,000 images on an NVIDIA GeForce RTX 2080 Ti, roughly $492\times$ the wall-clock cost of a ResNet-18 forward pass on the same hardware. Section 6.3 shows that $K = 10$ reduces this to 972.9 s (about 16.2 minutes; $68.6\times$ ResNet-18 cost) while achieving 98.2% AUROC, close to the 98.6% at $K = 50$.

Threshold Calibration: The threshold τ is calibrated on a validation set to achieve a target false positive rate (typically 5%). We set τ to the $(1 - \text{FPR}_{\text{target}})$ -th quantile of s_{OOD} scores computed on ID validation samples.

3.1.4. Evaluation Metrics

Following standard practice in OOD detection, we evaluate our methods using three primary metrics:

Area Under the ROC Curve (AUROC): The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) as the threshold τ varies:

$$\text{TPR}(\tau) = P_{x \sim P_{\text{ood}}}[s_{\text{OOD}}(x) > \tau] \quad (3.6)$$

$$\text{FPR}(\tau) = P_{x \sim P_{\text{id}}}[s_{\text{OOD}}(x) > \tau] \quad (3.7)$$

The AUROC measures the probability that a randomly chosen OOD sample has a higher score than a randomly chosen ID sample:

$$\text{AUROC} = P_{x_{\text{OOD}} \sim P_{\text{ood}}, x_{\text{ID}} \sim P_{\text{id}}}[s_{\text{OOD}}(x_{\text{OOD}}) > s_{\text{OOD}}(x_{\text{ID}})] \quad (3.8)$$

An AUROC of 1.0 indicates perfect separation, while 0.5 indicates random guessing. Higher values are better.

False Positive Rate at 95% TPR (FPR@95%TPR): This metric measures the fraction of ID samples incorrectly flagged as OOD when the threshold is set such that 95% of OOD samples are correctly detected:

$$\text{FPR@95\%TPR} = \text{FPR}(\tau^*) \quad \text{where} \quad \tau^* = \sup\{\tau : \text{TPR}(\tau) \geq 0.95\} \quad (3.9)$$

This metric is particularly important for safety-critical applications where we must detect most OOD samples while minimising false alarms. Lower values are better.

Balanced Detection Accuracy: The balanced accuracy at an optimal threshold (chosen to maximise Youden's J statistic):

$$\text{Balanced Accuracy} = \max_{\tau} \left[\frac{1}{2} \text{TPR}(\tau) + \frac{1}{2} (1 - \text{FPR}(\tau)) \right] \quad (3.10)$$

We also report Area Under the Precision-Recall Curve (AUPRC) and Detection Error (average of false positive and false negative rates) for thorough evaluation.

3.2. Diffusion Models for OOD Detection

This section provides the theoretical foundation for using diffusion models for OOD detection and formalises why reconstruction error serves as an effective OOD signal.

3.2.1. Theoretical Foundation

Why Diffusion Models for OOD Detection? Diffusion models offer several advantages for OOD detection that address limitations of previous approaches.

First, they provide a reconstruction-based alternative to direct likelihood estimation. As discussed in Section 2.4, likelihood-based OOD detection can fail in high dimensions because likelihood is affected by both density and ambient volume. Instead of computing $p(x)$ directly, we measure reconstruction error, which acts as an empirical distance-like signal with respect to the learnt data support (a formal monotonic mapping to manifold distance is not guaranteed under arbitrary noise schedules and is left to future theoretical work). The denoising process can be viewed as a projection operator:

$$\pi_{\theta}(x) = \mathbb{E}[x_0 \mid x_T = x + \text{noise}] \quad (3.11)$$

where x_0 is the denoised reconstruction. The reconstruction error $\|x - \pi_{\theta}(x)\|$ provides a distance-like measure to the learnt manifold.

Second, unlike VAEs that decode in a single step, diffusion models iteratively refine over T steps, providing multiple opportunities to accumulate evidence about whether a sample fits the learnt distribution (Wyatt et al., 2022; Graham et al., 2023). Each denoising step $x_t \rightarrow x_{t-1}$ can be viewed as a gradient step towards high-density regions.

Third, diffusion models enjoy stable training: they avoid the mode collapse of generative adversarial networks (GANs) and the posterior collapse of VAEs, relying only on a simple mean squared error (MSE) objective.

Fourth, class-conditional density modelling allows the network to learn separate denoising pathways per class, effectively modelling $p_{\theta}(x \mid y)$ for each $y \in \mathcal{Y}$. This enables fine-grained OOD detection: we can measure the fit of a test sample to each known class individually.

3.2.2. Reconstruction Error as OOD Signal

We now formalise why reconstruction error from diffusion models provides an effective OOD signal.

Low-Dimensional Support Perspective: If class-conditional image data concentrate near lower-dimensional supports (Section 2.4), the conditional diffusion model learns to approximate these structures $\mathcal{M}_c$ for each class c . For an ID sample $x \sim p(x | y = c)$, the sample lies on or near $\mathcal{M}_c$ and the denoising process recovers x with low error. For an OOD sample $x \sim P_{\text{ood}}$, the sample lies far from all learnt supports, and denoising projects x towards the nearest learnt support, incurring high reconstruction error.

Score-Based Interpretation: Recall from Section 2.3 that the noise prediction network approximates the score function:

$$\epsilon_\theta(x_t, t, c) \approx -\sqrt{1 - \bar{\alpha}_t} \nabla_{x_t} \log p(x_t | y = c) \quad (3.12)$$

The reconstruction error in Equations 3.1–3.2 thus measures how well x aligns with the learnt score function for class c . Samples that require large score corrections (high error) are likely OOD.

Empirical Risk Minimisation Perspective: During training, the model minimises:

$$\mathcal{L}_c = \mathbb{E}_{x \sim p(x|y=c), t, \epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c \right) \right\|^2 \right] \quad (3.13)$$

For test samples x from $p(x | y = c)$, we expect $e_c(x) \approx \mathcal{L}_c$ (low, when c is the correct class condition for x). For OOD samples, there is a distribution mismatch, leading to $e_c(x) \gg \mathcal{L}_c$ (high).

Information-Theoretic View: The reconstruction error can be interpreted as a measure of the information content needed to correct the model’s predictions. ID samples require little correction information (they fit the model’s learnt patterns), while OOD samples require substantial correction (they deviate from learnt patterns).

Relation to the Likelihood Pitfall. The key difference from likelihood-based methods is that reconstruction error measures distance in the learnt representation space, not volume in the ambient space (Kirichenko et al., 2020; Fefferman et al., 2016). While a high-dimensional OOD manifold might have more ambient volume than a low-dimensional ID manifold (causing likelihood to fail), it cannot have systematically lower reconstruction error when trained only on ID data. The diffusion model’s denoising pathways are optimised specifically for ID manifolds, making OOD samples difficult to reconstruct accurately.

3.2.3. Heuristic Analysis of Reconstruction Error Separation

We now provide a heuristic analysis establishing conditions under which reconstruction error separates ID from OOD samples. The following statements are motivational rather than fully proved: Informal Argument 1 is a standard bias–variance decomposition and Informal Claim 2 is grounded in the training objective. A rigorous treatment would require explicit assumptions on support geometry, noise-schedule dependence, and Lipschitz regularity of the score function, which we leave as future theoretical work.

Setup. Let $\mathcal{M}_{\text{ID}} \subset \mathbb{R}^D$ denote the support of the in-distribution, and let $\epsilon^*(x_t, t, c)$ denote the Bayes-optimal denoiser for class c , which minimises the expected MSE over the class-conditional distribution. The trained model ϵ_θ approximates ϵ^* with approximation error $\delta_\theta = \mathbb{E}[\|\epsilon_\theta - \epsilon^*\|^2]$.

Informal Argument 1 (*Standard MSE decomposition; ID reconstruction error bound*). Assuming x is drawn i.i.d. from the class-conditional training distribution, the Bayes-optimal denoiser is evaluated under the same noising process as training, and the learnt denoiser has approximation gap δ_θ , the expected reconstruction error under the correct class condition satisfies:

$$\mathbb{E}_{t,\epsilon} [\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2] \leq \mathcal{L}_c^* + \delta_\theta \quad (3.14)$$

where $\mathcal{L}_c^* = \mathbb{E}_{x \sim p(x|c), t, \epsilon} [\|\epsilon - \epsilon^*(x_t, t, c)\|^2]$ is the irreducible Bayes error for class c , and δ_θ is the model’s approximation gap.

Justification. This follows from the Pythagorean decomposition of MSE: for any estimator ϵ_θ , $\mathbb{E}[\|\epsilon - \epsilon_\theta\|^2] = \mathbb{E}[\|\epsilon - \epsilon^*\|^2] + \mathbb{E}[\|\epsilon^* - \epsilon_\theta\|^2]$. The Bayes-optimal denoiser achieves $\mathcal{L}_c^*$ on in-distribution data, and the learnt model adds at most δ_θ additional error due to finite capacity and training.

Informal Claim 2 (*OOD reconstruction error lower bound*). Assuming the OOD sample lies outside the support on which the class-conditional denoiser was optimised, the denoiser is locally regular rather than adversarially extrapolating, and the selected timestep range preserves information about the support mismatch, an OOD sample x_{ood} at support distance $d(x_{\text{ood}}, \mathcal{M}_c) = \inf_{x' \in \mathcal{M}_c} \|x_{\text{ood}} - x'\|$ satisfies:

$$\mathbb{E}_{t,\epsilon} [\|\epsilon - \epsilon_\theta(x_t^{\text{ood}}, t, c)\|^2] \geq \mathcal{L}_c^* + \Delta_{\text{ood}}(d) \quad (3.15)$$

where $\Delta_{\text{ood}}(d) > 0$ is a distribution mismatch term that is monotonically increasing in $d(x_{\text{ood}}, \mathcal{M}_c)$ for sufficiently large d .

Justification. The denoiser ϵ_θ is trained to minimise error over $p(x | c)$. When evaluated on $x_{\text{ood}} \notin \text{supp}(p(x | c))$, the noised input x_t^{ood} falls in a region of the noised space where the denoiser has not been optimised. For small noise levels ($t \ll T$), x_t^{ood} remains close to x_{ood} and thus far from the noised ID manifold, yielding large prediction error.

For large noise levels ($t \approx T$), both ID and OOD samples converge to isotropic Gaussian noise, reducing the gap. The expected mismatch Δ_{ood} thus depends on both the manifold distance d and the timestep distribution used for scoring.

Corollary (*Separation condition*). OOD detection via reconstruction error succeeds when:

$$\Delta_{\text{ood}}(d) > \delta_{\theta} + \text{Var}[e_{\text{ID}}] \quad (3.16)$$

where $\text{Var}[e_{\text{ID}}]$ is the variance of reconstruction error across ID samples. In words: the distribution mismatch must exceed the sum of model approximation error and natural ID variation.

Connection to Empirical Observations. This analysis explains several empirical findings:

- **Near-OOD vs. far-OOD detection difficulty:** In the reproducible core results, CIFAR-100 (a near-OOD dataset sharing low-level visual statistics with CIFAR-10) achieves higher AUROC (96.97%) than far-OOD datasets such as Textures (92.84%) and FashionMNIST (94.03%). This occurs because CIFAR-100’s airplane class is under-represented, creating a distributional gap that the model exploits effectively, which shows that semantic proximity does not deterministically predict detection difficulty.
- **Uniform timestep sampling is the best-tested strategy:** Averaging over all t captures both the high-sensitivity regime ($t \ll T$, where OOD deviation is maximal) and moderate-noise levels, maximising effective Δ_{ood} . Section 6.3 confirms that restricting to mid-range timesteps ($t \in [250, 750]$) reduces SVHN AUROC by 1.6%.
- **Separation loss improves both mean AUROC and stability:** The binary conditioning structure ($c=0$ for airplane, $c=1$ for all others) gives the model two class-conditional denoising paths. With binary conditioning but no separation loss ($\lambda = 0$), the three-seed Within-CIFAR mean is $92.52\% \pm 11.07\%$; adding the separation loss ($\lambda = 0.02$) raises the mean to $99.03\% \pm 0.07\%$ and largely removes the seed sensitivity.

Limitations. This analysis relies on several simplifying assumptions: (i) the ID distribution is concentrated near a learnable lower-dimensional support, (ii) the model’s approximation error δ_{θ} is small relative to the distribution mismatch, and (iii) the OOD distribution does not happen to lie in a region where the denoiser generalises well despite never training there. Assumption (iii) can fail for adversarially constructed OOD inputs. Formalising these bounds with explicit dependence on D , T , and support geometry remains an open theoretical challenge.

3.3. Model Architectures

We use a conditional diffusion model for OOD detection and then describe the multi-condition architecture used for the inkjet quality control application.

3.3.1. Conditional Diffusion Model

Our primary approach uses a conditional diffusion model that learns class-specific denoising processes.

Architecture: We employ a UNet-based architecture with the following components:

- **Encoder:** Downsampling path with residual blocks and self-attention at resolution 8×8
- **Bottleneck:** Multiple residual blocks with self-attention for capturing global context
- **Decoder:** Upsampling path with skip connections from encoder
- **Class Conditioning:** Learnt class embeddings $e_y \in \mathbb{R}^{d_{\text{emb}}}$ added to timestep embeddings
- **Timestep Embedding:** Sinusoidal positional encoding of timestep t

Mathematical Formulation: The network predicts noise conditioned on both timestep t and class y :

$$\epsilon_\theta(x_t, t, y) = \text{UNet}(x_t, \text{TimeEmbed}(t) + \text{ClassEmbed}(y)) \quad (3.17)$$

where $\text{TimeEmbed} : \{1, \dots, T\} \rightarrow \mathbb{R}^{d_{\text{emb}}}$ and $\text{ClassEmbed} : \{1, \dots, C\} \rightarrow \mathbb{R}^{d_{\text{emb}}}$ are learnt embedding functions.

Training: Following classifier-free guidance, we randomly drop the class conditioning with probability $p_{\text{uncond}} = 0.1$ during training:

$$\mathcal{L} = \mathbb{E}_{x, y, t, \epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, y' \right) \right\|^2 \right] \quad (3.18)$$

where $y' = y$ with probability $1 - p_{\text{uncond}}$ and $y' = \emptyset$ (null class) with probability p_{uncond} . This enables the model to perform both conditional and unconditional denoising.

OOD Detection: For a test sample x , we compute reconstruction error for each class:

$$\hat{e}_c(x) = \frac{1}{K} \sum_{k=1}^K \left\| \epsilon_k - \epsilon_\theta \left(\sqrt{\bar{\alpha}_{t_k}} x + \sqrt{1 - \bar{\alpha}_{t_k}} \epsilon_k, t_k, c \right) \right\|^2 \quad (3.19)$$

where we sample K random timesteps $t_k \sim \mathcal{U}[1, T]$ and noise vectors $\epsilon_k \sim \mathcal{N}(0, I)$ to obtain a Monte Carlo estimate of the population error $e_c(x)$. The OOD score is then computed by substituting $\hat{e}_0(x)$ and $\hat{e}_1(x)$ into Equation 3.3.

Design Rationale: The conditional model can learn class-specific patterns. ID samples from class c should be well-reconstructed using the c -conditioned denoising path but poorly reconstructed using other class paths. OOD detection therefore uses the contrastive score in Equation 3.3, comparing reconstruction errors under the ID and auxiliary non-ID conditions rather than relying on a minimum-error rule.

3.4. Inkjet Quality Control Pipeline

This section describes the two-stage pipeline for inkjet print quality classification using conditional diffusion models, which serves as a practical application of the generative classification principles developed above.

3.4.1. Two-Stage Pipeline Overview

Our pipeline processes full template images (1920×1080 pixels) of inkjet prints through two sequential stages: (1) feature detection using YOLOv8, and (2) quality classification using a conditional diffusion model (CDM).

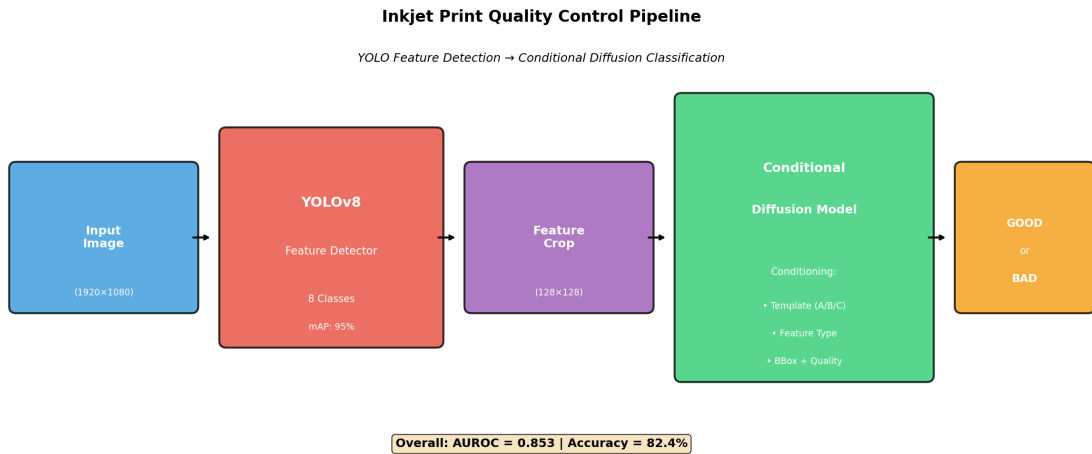


Figure 3.1.: Two-stage pipeline for inkjet print quality control. Stage 1: YOLOv8 detects and localises individual features on the template image (1920×1080). Stage 2: A conditional diffusion model classifies each extracted feature crop (128×128) as GOOD or BAD by comparing noise prediction errors under each quality condition.

Stage 1 (Feature Detection): A YOLOv8n model detects and localises 8 types of printed features (angle, dist1, dist6, dots, edge1–edge4) across three template types (A, B, C). Each detection provides a bounding box and feature class, which are used for cropping and conditioning.

Stage 2 (Quality Classification): Each detected feature is cropped, resized to 128×128 pixels, and classified by the CDM. The model compares noise prediction errors under GOOD and BAD quality conditions to determine the predicted class.

3.4.2. Dataset Overview

The public FTI_Zer0P inkjet dataset (Profactor, 2024) contains 1,204 inkjet print images from 15 building components, organised into 53 print datasets with up to three print areas per component. Of these images, 960 are labelled as GOOD or BAD for the relevant print-quality features. The retained CDM evaluation uses the labelled images after YOLO-based feature extraction: detections are grouped by 573 source images and expanded into 6,408 feature-level crops at 128×128 pixels across three template types, eight feature types, and binary quality labels (GOOD/BAD). Full dataset statistics, per-feature class balance, and the cross-validation protocol are provided in Section 5.5. All inkjet experiments in this thesis are evaluated exclusively on the public FTI_Zer0P dataset (Profactor, 2024), ensuring full external reproducibility.

3.4.3. Multi-Head Conditioning Mechanism

The quality classification CDM extends the standard class-conditional diffusion model with a multi-head conditioning mechanism that incorporates four types of metadata simultaneously.

Conditioning Inputs: The model conditions on:

1. **Template type** (categorical, 3 classes): learnt embedding $e_{\text{tpl}} \in \mathbb{R}^{256}$ implemented as $\text{Embed}(3, 256)$.
2. **Feature type** (categorical, 8 classes): learnt embedding $e_{\text{feat}} \in \mathbb{R}^{256}$ implemented as $\text{Embed}(8, 256)$.
3. **Quality label** (binary, GOOD/BAD): learnt embedding $e_{\text{qual}} \in \mathbb{R}^{256}$ implemented as $\text{Embed}(2, 256)$.
4. **Bounding box** (continuous, 4 values): MLP-encoded embedding $e_{\text{bbox}} \in \mathbb{R}^{256}$ implemented as $\text{MLP}(4 \rightarrow 64 \rightarrow 256)$.

Combination: All conditioning embeddings in the inkjet pipeline are projected to the same dimensionality (256) and summed:

$$\mathbf{e}_{\text{cond}} = e_{\text{tpl}} + e_{\text{feat}} + e_{\text{qual}} + e_{\text{bbox}} + e_{\text{time}} \quad (3.20)$$

where e_{time} is the standard sinusoidal timestep embedding. The combined conditioning vector $\mathbf{e}_{\text{cond}}$ (distinct from the binary class variable $c \in \{0, 1\}$ used in the CIFAR-10 OOD formulation) is injected at each UNet resolution level.

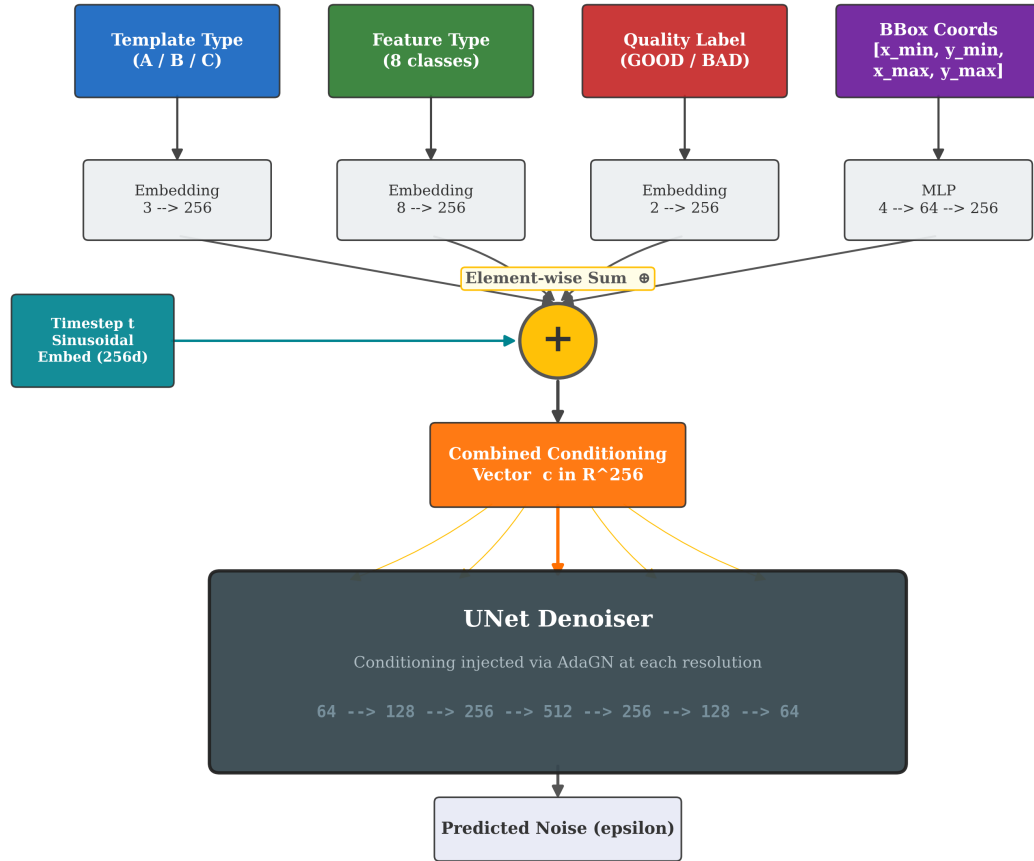


Figure 3.2.: Multi-head conditioning mechanism. Four conditioning heads, namely template type ($3 \rightarrow 256$), feature type ($8 \rightarrow 256$), quality label ($2 \rightarrow 256$), and bounding box coordinates ($4 \rightarrow 64 \rightarrow 256$), are independently encoded and combined via element-wise summation. The resulting conditioning vector is merged with the sinusoidal timestep embedding and injected into the UNet denoiser via Adaptive Group Normalisation (AdaGN) at each resolution level.

Noise Prediction: The network predicts noise conditioned on all metadata:

$$\epsilon_{\theta}(x_t, t, \text{tmpl}, \text{feat}, \text{qual}, \text{bbox}) = \text{UNet}(x_t, \mathbf{e}_{\text{cond}}) \quad (3.21)$$

3.4.4. Quality Classification via Differential Noise Prediction

Classification leverages the CDM’s conditional noise prediction to compare how well the model reconstructs each sample under GOOD vs. BAD quality assumptions.

Algorithm 2 *Quality Classification via Differential Noise Prediction.*

Require: Feature crop $x \in \mathbb{R}^{3 \times 128 \times 128}$, trained CDM ϵ_θ
Require: Feature metadata: tmpl , feat , bbox
Require: Number of Monte Carlo trials K , number of timesteps T
Ensure: Predicted quality $\hat{q} \in \{\text{GOOD}, \text{BAD}\}$, confidence
Notation: $c_g \equiv (\text{tmpl}, \text{feat}, \text{GOOD}, \text{bbox})$; $c_b \equiv (\text{tmpl}, \text{feat}, \text{BAD}, \text{bbox})$.
 1: $e_g \leftarrow 0$; $e_b \leftarrow 0$
 2: **for** trial $k = 1$ to K **do**
 3: $t_k \sim \text{Uniform}\{1, \dots, T\}$; $\epsilon_k \sim \mathcal{N}(0, I)$
 4: $x_{t_k} \leftarrow \sqrt{\alpha_{t_k}} x + \sqrt{1 - \alpha_{t_k}} \epsilon_k$
 5: $e_g \leftarrow e_g + \|\epsilon_k - \epsilon_\theta(x_{t_k}, t_k, c_g)\|^2$
 6: $e_b \leftarrow e_b + \|\epsilon_k - \epsilon_\theta(x_{t_k}, t_k, c_b)\|^2$
 7: **end for**
 8: $e_g \leftarrow e_g / K$; $e_b \leftarrow e_b / K$
 9: **if** $e_g < e_b$ **then** $\hat{q} \leftarrow \text{GOOD}$ **else** $\hat{q} \leftarrow \text{BAD}$
 10: confidence $\leftarrow |e_g - e_b| / (e_g + e_b)$; **return** $\hat{q}$, confidence

Key Difference from OOD Detection: In the OOD detection formulation (Algorithm 1), we use a binary conditional setup ($C = 2$: $c = 0$ for ID, $c = 1$ for OOD), so scoring reduces to a single contrastive comparison via $O(2K)$ forward passes. In quality classification, we similarly compare only two conditions (GOOD vs. BAD) while keeping all other conditions (template, feature, bbox) fixed, yielding the same $O(2K)$ cost per sample.

Polarity alignment across tracks. The two experimental tracks inherit opposite native label conventions: the CIFAR-10 OOD track encodes $y = 1$ as the OOD-proxy class (OOD-positive), while the public FTI_Zer0P inkjet dataset’s raw `metadata.csv` encodes `label = 1` as GOOD and `label = 0` as BAD (ID-positive). To keep the OOD-positive scoring convention stated in Section 2.1.1 consistent across both tracks, the inkjet pipeline inverts the raw label once at the dataset boundary, `quality = 1 - label`, so that the model is trained with $c = 0$ for GOOD (ID) and $c = 1$ for BAD (OOD-proxy), matching the CIFAR track. The inkjet anomaly score is therefore $s_{\text{QC}}(x) = e_g - e_b = e_{c=0} - e_{c=1}$, with higher values indicating *more anomalous* (BAD-like) crops, and AUROC is reported with BAD as the positive class. With this alignment every AUROC and FPR95 reported in this thesis follows the same direction (higher score $\Rightarrow$ stronger evidence of being out-of-distribution or defective), independent of the underlying dataset’s label dictionary.

3.5. Training Strategies

Our training procedures incorporate specific loss functions, optimisation strategies, and data augmentation techniques.

3.5.1. Loss Functions and Optimisation

Primary Loss: We use the standard DDPM training objective:

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{x,y,t \sim \mathcal{U}[1,T], \epsilon \sim \mathcal{N}(0,I)} \left[\left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, y \right) \right\|^2 \right] \quad (3.22)$$

This simple mean squared error (MSE) loss has proven highly effective for training diffusion models.

Separation Loss: To explicitly encourage the model to produce distinct class-conditional noise predictions (and thereby distinct reconstruction error distributions) for ID and OOD conditions, we introduce a contrastive push-apart loss:

$$\mathcal{L}_{\text{sep}}(x_t, t) = - \left\| \epsilon_{\theta}(x_t, t, c=0) - \epsilon_{\theta}(x_t, t, c=1) \right\|^2 \quad (3.23)$$

where $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon$ is the noised image obtained via the same forward-diffusion process used by $\mathcal{L}_{\text{diffusion}}$ above, and $\epsilon_{\theta}(x_t, t, c)$ is the noise predicted by the UNet under class condition c . Minimising this term (as part of the total loss) maximises the squared Euclidean distance between class-conditional noise predictions, directly encouraging the model to distinguish ID from OOD inputs at every denoising step. Computing $\mathcal{L}_{\text{sep}}$ requires two additional forward passes of the UNet (one with $c=0$ and one with $c=1$) beyond the single forward pass already needed for $\mathcal{L}_{\text{diffusion}}$, so each training step performs three forward passes when $\lambda > 0$. In practice this triples the per-step training compute relative to the $\lambda=0$ baseline; the explicit per-epoch cost is quantified in Section 7.5.

The total training objective combines the diffusion loss with the separation loss:

$$\mathcal{L} = \mathcal{L}_{\text{diffusion}} + \lambda \cdot \mathcal{L}_{\text{sep}} \quad (3.24)$$

where $\lambda \geq 0$ controls the strength of the separation objective. The ablation study in Section 6.3 systematically evaluates $\lambda \in \{0.0, 0.001, 0.01, 0.02, 0.05, 0.1\}$; the final selected CIFAR-10 model uses $\lambda = 0.02$, while the K , timestep-sampling, and scoring-strategy ablations use the seed-42 $\lambda = 0.01$ checkpoint for compute-budget reasons.

Relation to Contrastive Objectives. The separation loss is structurally related to contrastive training objectives used in representation learning for OOD detection (Tack et

al., 2020; Wu et al., 2024). However, $\mathcal{L}_{\text{sep}}$ differs in two key respects: (i) it operates in noise-prediction space rather than representation space, directly maximising the gap between class-conditional denoising outputs; and (ii) it pushes apart predictions from a single shared UNet under different conditioning inputs, rather than training separate encoders or using data augmentation to create contrastive pairs. This design choice is specific to the conditional diffusion framework and enables the separation signal to be injected at the same granularity (per-timestep, per-pixel) as the primary denoising objective.

Because $\mathcal{L}_{\text{sep}}$ is a negative squared norm, it is not bounded below in isolation. The method therefore relies on the denoising objective, small tuned values of λ , and empirical validation to prevent unproductive norm growth in the two conditional predictions. This risk is one reason we treat λ as an ablated hyperparameter rather than a fixed theoretical constant; empirically, the high-weight $\lambda = 0.1$ setting drops to 96.67% Within-CIFAR AUROC, consistent with the separation objective beginning to dominate the denoising loss.

Noise Schedule: For CIFAR-10 experiments, we use a cosine variance schedule as it maintains more signal at higher timesteps compared to linear schedules:

$$\bar{\alpha}_t = \cos^2\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right) \quad (3.25)$$

where $s = 0.008$ is a small offset parameter. The retained public inkjet pipeline also uses a cosine noise schedule with $T = 1000$ timesteps, matching the implementation summarised in Chapter 5 and Appendix C.

3.5.2. Training Procedure

During each training iteration, a batch of images is noised to random timesteps $t \sim \mathcal{U}[1, T]$, the conditional UNet predicts the added noise, and the combined loss $\mathcal{L} = \mathcal{L}_{\text{diffusion}} + \lambda \mathcal{L}_{\text{sep}}$ (Equation 3.24) is minimised via AdamW (Loshchilov and Hutter, 2019) with cosine-annealed learning rate ($\eta = 10^{-4}$, weight decay 0.01, 5-epoch warm-up). An exponential moving average (EMA) of model parameters with decay $\gamma = 0.9999$ is maintained for evaluation. Full training configurations and hyperparameter tables are provided in Chapter 5 and Appendix C.

3.5.3. Data Preprocessing

All images are normalised to $[-1, 1]$ before being passed to the diffusion model. CIFAR-10 images are augmented during training with horizontal flips and random crops (4-pixel padding). Inkjet crops are resized to 128×128 with no augmentation. Class balancing uses weighted sampling with $w_c = 1/(n_c \cdot C)$. Full preprocessing and augmentation details are provided in Chapter 5.

3.6. Evaluation Framework

We evaluate using AUROC (probability that a random OOD sample scores higher than a random ID sample; 1.0 = perfect, 0.5 = random) and FPR@95%TPR (false positive rate at 95% true positive rate; lower is better) as primary metrics, with AUPRC and Detection Error reported for completeness. Formal definitions are provided in Section 3.1.4.

The threshold τ is set using Youden’s J statistic $\tau^* = \arg \max_{\tau} [\text{TPR}(\tau) - \text{FPR}(\tau)]$ for balanced scenarios, or at a fixed FPR quantile when controlling false alarm rates. CIFAR-10 reports multi-seed descriptive summaries only for the $\lambda \in \{0.01, 0.02\}$ separation-loss checkpoints; other CIFAR ablations use the documented seed-42 run. Inkjet results use paired t -tests and Wilcoxon signed-rank tests with Holm correction over 5 folds.

All models are implemented in PyTorch Lightning with Hydra-based configuration; full implementation details are covered in Chapter 4.

4. Implementation

We implement the theoretical framework from Chapter 3 as a modular research prototype using modern deep learning frameworks. The implementation covers two experimental tracks: (1) diffusion-based OOD detection on CIFAR-10, and (2) the inkjet print quality classification pipeline. Our architecture emphasises modularity, traceability, and extensibility. The implementation leverages PyTorch Lightning (Falcon and team, 2019) for training orchestration, the Hugging Face Diffusers library for diffusion model components, and Hydra (Yadan, 2019) for flexible configuration management. All code follows standard practices for documentation and verifiability.

4.1. System Overview and Design

Our implementation consists of five core components: (1) configuration management (Hydra), (2) data loading (PyTorch Lightning DataModule), (3) model definitions (binary conditional diffusion with UNet), (4) training orchestration (Lightning Trainer), and (5) evaluation framework (metrics and OOD detection).

The system design prioritises three principles: **modularity** (independent components with clear interfaces enabling easy extension), **configurability** (all hyperparameters externalised in YAML files enabling traceable experimentation without code modification), and **reproducibility** (deterministic training through seed management and complete configuration logging). Models inherit from a common base class implementing shared functionality (noise scheduling, loss computation, optimisation), while specific variants (binary conditional UNet for OOD detection, multi-head conditional CDM for inkjet QC) override methods for task-specific behaviour. This inheritance-based design eliminates code duplication while preserving flexibility.

The Hydra configuration system follows hierarchical composition, where base configurations define defaults that can be overridden by experiment-specific settings. For example, a conditional diffusion UNet model can be launched with default settings via `python train.py`, or customised with: `python train.py model.learning_rate=1e-4 trainer.devices=4`. Configuration overrides are automatically logged and saved alongside results.

4.2. CIFAR-10 OOD Detection Implementation

4.2.1. Model Architecture Implementations

Conditional Diffusion with UNet The primary model uses the UNet2DModel from the Hugging Face Diffusers library with the following configuration: 3 input channels (RGB), block_out_channels=(128, 256, 256, 512) (base channel 128 with multipliers [1, 2, 2, 4]), attention at resolutions [16, 8], 2 residual blocks per resolution, and 512-dim embeddings for both timestep and class (in contrast to the 256-dim embeddings used in the inkjet pipeline). Class information is incorporated through learnt embeddings injected via adaptive group normalisation (AdaGN) layers at multiple scales.

Training uses AdamW optimisation (learning rate 10^{-4} , weight decay $\lambda_{wd} = 0.01$) with cosine annealing and a 5-epoch linear warm-up, effective batch size 128 (batch 64 with gradient accumulation $\times 2$), and up to 200 epochs on CIFAR-10 (early stopping, patience=30 on validation AUROC; typical convergence at epoch 15–25). For the binary CDM experiments, conditioning is reduced to two classes (airplane as ID, remaining nine classes as OOD), yielding a model of approximately 68.8M parameters (verified by direct parameter count of the instantiated ConditionalUNet using the training-run configuration logged in W&B).

4.2.2. CIFAR-10 Data Pipeline

Data loading uses PyTorch Lightning’s LightningDataModule abstraction enabling seamless dataset switching. CIFAR-10 is split into 45K training, 5K validation, and 10K test sets with standard augmentation (horizontal flip). The retained external OOD datasets (CIFAR-100, Places365, FashionMNIST, SVHN, Textures) undergo identical preprocessing (normalisation to $[-1, 1]$) without augmentation during evaluation. FashionMNIST images are resized from 28×28 to 32×32 and converted from greyscale to 3-channel RGB.

4.2.3. OOD Evaluation Pipeline

Evaluation computes reconstruction error by sampling K random timesteps for each test image, following Algorithm 3.1.3 in Chapter 3. All metrics (AUROC, FPR@95%TPR, accuracy) use scikit-learn implementations. W&B logging tracks metrics per epoch, and model checkpoints are saved at best validation loss and best OOD AUROC.

4.3. Inkjet Quality Control Implementation

The inkjet quality control pipeline extends the CIFAR-10 OOD framework to a practical industrial setting. This section describes the implementation of each pipeline com-

ponent: the YOLOv8 feature detector, the multi-condition CDM, and the comparison methods.

4.3.1. YOLOv8 Feature Detector

Model Selection We use YOLOv8n (nano), the smallest variant with $\sim 3.2\text{M}$ parameters, which provides sufficient detection accuracy for our well-structured template images while enabling fast inference. The model is trained on the thesis feature annotations and accessed through the Ultralytics library.

Training Configuration

- Input resolution: 640×640 (standard YOLO input)
- Epochs: 100 with early stopping (patience 20)
- Optimiser: SGD with momentum 0.937, weight decay 0.0005
- Learning rate: 0.01 with cosine annealing
- Augmentation: mosaic, mixup, flip, HSV colour jitter

Detection Output For each template image, the detector outputs bounding boxes $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$, class predictions (8 feature types), and confidence scores. Only detections with confidence > 0.3 are retained. The expected number of detections per template ranges from 2 to 5 depending on the template type, yielding an average of ~ 4.8 features per image across the dataset.

Feature Extraction Each detected region is cropped from the original 1920×1080 image and resized to 128×128 pixels using bilinear interpolation. The crop coordinates, feature class, and template type are stored as metadata for downstream CDM conditioning.

4.3.2. Conditional Diffusion Model for Quality Classification

Architecture The inkjet CDM uses a custom NoisePredictorV3 UNet (not UNet2DModel) tailored for 128×128 input with multi-head conditioning:

- Input channels: 3 (RGB)
- Base channels: 64
- Channel multipliers: $[1, 2, 4, 8]$ (effective block channels 64, 128, 256, 512)
- Attention at resolutions: $[32, 16]$

- Residual blocks per level: 2
- Total parameters: $\sim 9.3\text{M}$ (verified by direct parameter count of the NoisePredictorV3 module with `base_channels=64`)

Multi-Head Conditioning Implementation The four conditioning heads (Section 3.4.3) are implemented as follows:

- Template embedding: `nn.Embedding(3, 256)` encodes template type (A/B/C) into a 256-dim vector
- Feature embedding: `nn.Embedding(8, 256)` encodes feature type into a 256-dim vector
- Quality embedding: `nn.Embedding(2, 256)` encodes the GOOD/BAD label into a 256-dim vector
- Bbox encoder: `nn.Sequential(Linear(4, 64), GELU, Linear(64, 256))` encodes normalised bounding box coordinates
- Timestep embedding: standard sinusoidal encoding $\rightarrow$ MLP $\rightarrow$ 256-dim vector

All five embeddings are summed to produce the final conditioning vector $c \in \mathbb{R}^{256}$, which is injected via AdaGN at each UNet resolution level.

Training Details

- Optimiser: AdamW ($\text{lr} = 2 \times 10^{-4}$, weight decay = 10^{-4})
- Batch size: 128 for the $\lambda = 0$ baseline and 64 for $\lambda > 0$ runs with separation loss
- Noise schedule: Cosine, $T = 1000$
- Training: 100 epochs per fold
- Hardware: Single GPU per run (V100 16 GB, GV100 32 GB, or P40 24 GB depending on availability); training time $\sim 6\text{--}8\text{h}$ per fold on V100, comparable on GV100/P40

Inference Quality classification follows Algorithm 3.4.4. The public implementation defaults to $K = 50$ Monte Carlo trials per sample, while the retained 5-fold cross-validation ablation reported in Chapter 6 uses $K = 100$ for more stable estimates. For each trial, a random timestep $t \sim \mathcal{U}[1, 1000]$ and noise $\epsilon \sim \mathcal{N}(0, I)$ are sampled. The model predicts noise under both GOOD and BAD conditions, and the class with lower reconstruction error is selected as the prediction.

4.3.3. Comparison Method Implementations

Retained public comparison scope. Earlier internal experiments considered supervised and anomaly-detection baselines, but those non-CDM pipelines are not retained as thesis evidence because their full public code, splits, configurations, and result artefacts are not part of the released package. The implementation described here therefore focuses on the public YOLO+CDM pipeline that can be reproduced from Inkjet00D, the public FTI_Zer0P dataset, and the released inkjet CDM weights.

4.3.4. Public Inkjet Evaluation Infrastructure

The public inkjet implementation retained in this thesis is the Inkjet00D CDM pipeline together with its training, evaluation, ablation, and 5-fold cross-validation scripts.

Public CDM Components

- **Training entry point:** `train.py` reproduces the multi-head CDM training run for a chosen separation-loss weight.
- **Evaluation entry point:** `evaluate.py` reproduces GOOD-vs.-BAD scoring with configurable K Monte Carlo trials.
- **Ablation script:** `run_ablation.py` evaluates $\lambda \in \{0.0, 0.01, 0.02, 0.05\}$ under a shared train/test split.
- **Cross-validation script:** `run_cv.py` runs the retained 5-fold image-level protocol and writes fold summaries used in Chapter 6.

4.3.5. Cross-Validation Infrastructure

Stratified 5-Fold Cross-Validation The retained public CDM experiments use image-level splitting rather than sample-level splitting, preventing data leakage from correlated crops derived from the same template image.

Splitting Algorithm

1. Group samples by processed source image (573 groups)
2. Stratify groups by quality label distribution
3. Split groups into 5 folds with balanced label ratios
4. Expand folds to sample level for training and evaluation

This ensures that each fold has a representative distribution of all template types, feature types, and quality labels.

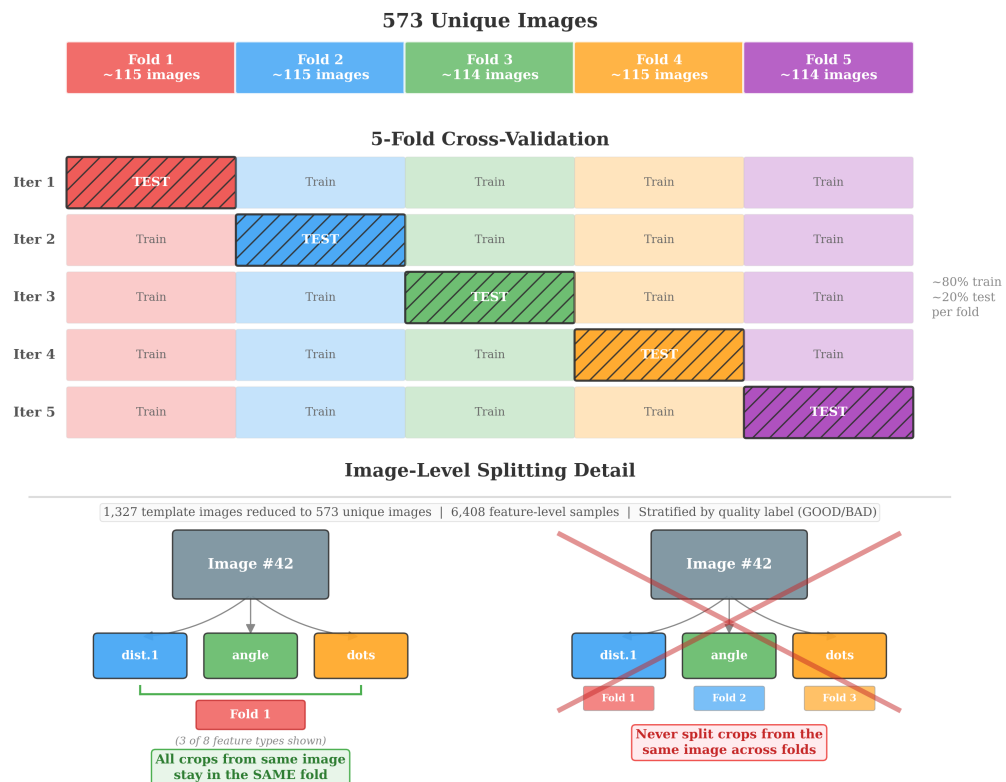


Figure 4.1.: Stratified 5-fold cross-validation with image-level splitting. The 573 processed source-image groups are partitioned into 5 folds (114–115 groups per fold), with each fold serving as the test set exactly once. Bottom: image-level splitting ensures all feature crops derived from the same source image remain in the same fold, preventing data leakage between training and test sets.

4.4. Key Engineering Considerations

Memory Efficiency Training across the available GPU pool (V100 16 GB, GV100 32 GB, P40 24 GB) requires careful memory management. We use mixed-precision (AMP) training, reducing peak activation memory to $\approx 4\text{--}6$ GB and enabling execution even on the V100 (16 GB, the tightest available GPU), enable gradient accumulation to maintain effective batch sizes with smaller per-GPU batches, and implement efficient data loading with 8 worker processes and pinned memory.

Inference Cost OOD detection requires K forward passes per image. On an NVIDIA GeForce RTX 2080 Ti, $K = 50$ takes 4861.1 s per 10K images ($492\times$ ResNet-18 wall-clock cost); $K = 10$ takes 972.9 s ($68.6\times$) while maintaining 98.2% AUROC. Both timings use the seed-42 $\lambda=0.01$ checkpoint; $K = 1$ achieves 91.0% AUROC but was not re-timed, so no wall-clock ratio is claimed. For the inkjet pipeline, $K = 50$ takes ~ 0.5 seconds per feature crop.

Determinism and Logging All retained experiments use fixed random seeds with PyTorch deterministic mode enabled. Configuration files, git commit hashes, and environment metadata (Python/CUDA versions) are logged automatically. Public verification draws on the released repositories, weights, raw scores, configuration files, and command sequences detailed in Appendix C.

Multi-GPU Training The code supports PyTorch Lightning’s DDP with automatic gradient synchronisation, but all reported experiments use a single GPU per run; multi-GPU support is an implementation capability rather than part of the experimental protocol.

4.5. Reference Implementation

CL-2 Reproducibility. The CIFAR-10/OOD implementation is publicly available at <https://github.com/ahmed-3m/DiffusionOOD>, with weights at <https://huggingface.co/ahmed-3m/DiffusionOOD>. The inkjet CDM implementation, experiment documentation, and figure-generation scripts are publicly available at <https://github.com/ahmed-3m/InkjetOOD> (Mohammed, 2026b), with weights at <https://huggingface.co/ahmed-3m/InkjetOOD>; the accompanying FTI_Zer0P dataset is available from Zenodo at DOI 10.5281/zenodo.11444566 (Profactor, 2024). The codebases follow standard PyTorch (Paszke et al., 2019) project layouts with modules for models, data loading, evaluation, training scripts, and configuration. Key software dependencies and versions are listed in Appendix C.6.

5. Experimental Setup

This chapter describes the experimental design for both evaluation tracks, answering the research questions posed in Chapter 1. Sections 5.1–5.4 cover the CIFAR-10 track; Section 5.5 covers the inkjet pipeline.

5.1. Datasets

5.1.1. In-Distribution Dataset: CIFAR-10

Dataset description. CIFAR-10 (Krizhevsky, Hinton, et al., 2009) consists of 60,000 32×32 colour images across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Binary split. We adopt a binary framing: the airplane class (class 0) forms the in-distribution set, and the remaining nine classes form the OOD proxy. This yields a controlled binary OOD problem where the ID/OOD boundary is known.

Data splits.

- Training set: 45,000 images (90% of original 50,000 training images, stratified)
- Validation set: 5,000 images (10% of training, stratified)
- Test set: 10,000 images (standard CIFAR-10 test set)

Preprocessing. All images are converted to PyTorch tensors, normalised to $[-1, 1]$ via $x' = 2x - 1$, and augmented during training with random horizontal flips. Validation and test sets receive only normalisation.

We evaluate on five public external OOD datasets spanning near- to far-OOD distributional distances from CIFAR-10.

Near-OOD: CIFAR-100 (Krizhevsky, Hinton, et al., 2009) (10,000 test images, 32×32 , 100 classes sharing CIFAR-10 image statistics).

Far-OOD: SVHN (Netzer et al., 2011) (26,032 digit images, 32×32); **FashionMNIST** (Xiao et al., 2017) (10,000 greyscale clothing images, resized to 32×32 , converted to 3-channel); **Textures (DTD)** (Cimpoi et al., 2014) (1,880 texture images, cropped and resized to 32×32); **Places365** (Zhou et al., 2018) (10,000 scene images, 32×32). All datasets are normalised to $[-1, 1]$.

5.1.2. Dataset Statistics

Table 5.1.: Dataset statistics for in-distribution and out-of-distribution datasets, spanning near-OOD (shared visual statistics) to far-OOD (distinct domains) evaluation scenarios.

Dataset	Type	Test Images	Classes	Resolution
CIFAR-10 (airplane)	ID	1,000	1	32×32
CIFAR-10 (rest)	OOD proxy	9,000	9	32×32
CIFAR-100	Near OOD	10,000	100	32×32
Places365	Far OOD	10,000	365	32×32
SVHN	Far OOD	26,032	10	32×32
FashionMNIST	Far OOD	10,000	10	28×28
Textures (DTD)	Far OOD	1,880	47	Variable

5.2. Experimental Design

5.2.1. Primary Experiment: Binary CDM Robustness Across Seeds

We train the binary conditional diffusion model with three random seeds ($\{42, 123, 456\}$) for the primary reported separation-loss settings. The final selected model uses $\lambda = 0.02$; $\lambda = 0.01$ is retained for the K , timestep-sampling, and scoring-strategy ablations for compute-budget reasons. The three seeds control model initialisation, data shuffling, and all stochastic training operations.

Research question: Is the binary CDM’s OOD detection performance stable across training runs, and what is its mean AUROC and standard deviation on the within-CIFAR binary test split?

Evaluation: Within-CIFAR AUROC and FPR@95%TPR are the primary metrics. For reproducibility, all reported core external-OOD claims use public datasets, released code, and preserved raw-score artefacts (seed-42 raw-score evaluation; $K = 100$ MC trials).

5.2.2. Separation Loss Ablation

We conduct a complete sweep over six separation loss weights: $\lambda \in \{0.0, 0.001, 0.01, 0.02, 0.05, 0.1\}$. The reference ablation checkpoint $\lambda = 0.01$ is evaluated with three seeds (seeds 42/123/456); the final selected weight $\lambda = 0.02$ is also evaluated with three seeds. All other λ values are evaluated with seed 42 only. Each λ value is additionally evaluated on SVHN to assess cross-domain stability.

Research question: How much does the separation loss improve OOD detection, and what is the best-tested weight λ ?

5.2.3. Ablation 1: Number of Monte Carlo Trials (K)

We vary the number of timestep samples $K \in \{1, 5, 10, 25, 50, 100\}$ used to compute the OOD score (seed-42 checkpoint, $\lambda = 0.01$).

Research question: What is the accuracy–efficiency trade-off as K increases, and what is the minimum K for near-optimal performance?

Evaluation: AUROC, FPR@95%TPR, and wall-clock inference time (seconds per 10K images on an NVIDIA GeForce RTX 2080 Ti for the runtime table).

5.2.4. Ablation 2: Timestep Sampling Strategy

We compare three strategies for sampling timesteps during OOD scoring (seed-42 checkpoint, $\lambda = 0.01$, $K = 50$):

- **Uniform:** $t \sim \mathcal{U}[1, T]$ uniformly
- **Mid-focus:** Truncated normal $t \sim \mathcal{N}_{\text{trunc}}(\mu = 300, \sigma = 150)$
- **Stratified:** Divide $[1, T]$ into K equal bins; sample one t per bin

Research question: Which noise-level coverage is most informative for OOD detection?

5.2.5. Ablation 3: OOD Scoring Method

We compare three scoring formulations (seed-42 checkpoint, $\lambda = 0.01$, $K = 50$):

- **Difference:** $s(x) = e_0(x) - e_1(x)$ (ID error minus OOD error)

- Ratio: $s(x) = e_0(x)/e_1(x)$ (ID error divided by OOD error)
- **ID error only:** $s(x) = e_0(x)$ (no contrastive comparison)

Research question: Is contrastive scoring (using both binary conditions) essential, and do difference and ratio scoring differ significantly?

5.3. Implementation Details

5.3.1. Binary CDM Architecture

The binary CDM uses a UNet backbone with base channel width 128, channel multipliers $[1, 2, 2, 4]$ (yielding channels $[128, 256, 256, 512]$), attention at resolutions 16×16 and 8×8 , two residual blocks per resolution, and class-conditional Adaptive Group Normalisation (AdaGN); full architectural details are provided in Chapter 4 and Appendix C. The model has ~ 68.8 M trainable parameters with binary conditioning ($c = 0$: airplane, $c = 1$: rest).

Noise schedule. Cosine schedule (A. Q. Nichol and Dhariwal, 2021) with $T = 1000$ timesteps: $\bar{\alpha}_t = \cos^2\left(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\right)$, $s = 0.008$.

Training configuration.

- Optimiser: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay $\lambda_{\text{wd}} = 0.01$)
- Learning rate: 10^{-4} with 5-epoch linear warm-up then cosine annealing
- Batch size: 128
- Classifier-free guidance dropout: $p_{\text{uncond}} = 0.1$
- Early stopping: patience 30 epochs on validation AUROC
- Hardware: Single GPU per run (V100 16 GB, GV100 32 GB, or P40 24 GB); training time ≈ 10 –16 h on GV100, ~ 18 –28 h on V100/P40

OOD scoring. Default: $K = 50$ MC trials, uniform timestep sampling, difference scoring. External OOD evaluation uses $K = 100$ for higher stability.

Separation loss. During training, the separation loss $\mathcal{L}_{\text{sep}}$ (Equation 3.23) is added with weight λ as part of the total training objective (Equation 3.24). The separation loss is computed on each mini-batch using the predicted noise errors under both conditions.

5.3.2. Inkjet CDM Architecture

The inkjet CDM uses the same diffusion-scoring principle as the CIFAR model but a separate custom NoisePredictorV3 UNet tailored to 128×128 feature crops. It uses a multi-head conditioning mechanism:

- Template type embedding (3 values: A, B, C)
- Feature type embedding (8 values: dist1, dist6, edge1–edge4, angle, dots)
- Quality label embedding (2 values: GOOD, BAD)
- Bounding box coordinates (4 values, normalised)

Input resolution: 128×128 (YOLO-cropped feature patches). Training uses stratified 5-fold CV with image-level splitting. The public implementation defaults to $K = 50$ inference trials, while the retained separation-loss ablation reported in Chapter 6 uses $K = 100$ to reduce Monte Carlo scoring variance.

5.4. Evaluation Protocol

5.4.1. Metrics

Primary metrics: AUROC (overall discrimination; 1.0 = perfect, 0.5 = random) and FPR@95%TPR (false positive rate at 95% true positive rate; lower is better). Formal definitions, including the TPR/FPR equations, are provided in Section 3.1.4; AUPRC and Detection Error are reported for completeness.

Reporting: Seed-level CIFAR analyses with $n < 5$ are reported descriptively (no inferential significance claims). For inkjet ($n = 5$ folds), significance uses paired t -test and Wilcoxon signed-rank tests with Holm correction.

5.4.2. Within-CIFAR Evaluation

For the within-CIFAR binary test split: ID samples are the 1,000 CIFAR-10 test images of the airplane class. OOD samples are the 9,000 CIFAR-10 test images of the nine remaining classes. AUROC is computed using scikit-learn with default parameters.

5.4.3. External OOD Evaluation

For each external OOD dataset: the reference pool is the full CIFAR-10 test set (10,000 images), because the binary CDM was trained on both conditional branches of CIFAR-10: airplane as $c=0$ and the remaining nine classes as $c=1$. OOD samples are the full external dataset test split. Scores are computed with $K = 100$ MC trials. AUROC and

FPR@95%TPR are reported for the retained reproducible raw-score runs. Importantly, no external OOD data was used for threshold calibration, hyperparameter selection, or early stopping; all model selection was performed on the within-CIFAR validation split (Section 5.4.2).

5.4.4. Reproducibility

All experiments use deterministic PyTorch settings:

```
torch.backends.cudnn.deterministic = True
torch.use_deterministic_algorithms(True)
```

Each checkpoint saves model state, EMA state, optimiser state, and training step. The best validation AUROC checkpoint is used for all evaluations. Training and evaluation scripts for the CIFAR-10 experiments are available at <https://github.com/ahmed-3m/DiffusionOOD>; pre-trained model weights for the retained seed and λ configurations are hosted at <https://huggingface.co/ahmed-3m/DiffusionOOD>, enabling direct evaluation without retraining. The inkjet CDM scripts are reproducible from the public <https://github.com/ahmed-3m/InkjetOOD> repository (Mohammed, 2026b) at commit 7c2451c; the corresponding pre-trained inkjet CDM weights are available at <https://huggingface.co/ahmed-3m/InkjetOOD>. The inkjet track also requires the public PROFACTOR FTI_Zer0P dataset on Zenodo (Profactor, 2024). After downloading the dataset, the expected layout is data/images/ for PNG files and data/metadata.csv with columns file_name, label, and feature. The repository documents training, evaluation, 5-fold cross-validation, and figure-regeneration commands for the retained inkjet CDM results. The corresponding public command sequence is:

```
git clone https://github.com/ahmed-3m/InkjetOOD.git
cd InkjetOOD
git checkout 7c2451c
pip install -r requirements.txt
python run_ablation.py --metadata ./data/metadata.csv --img_dir ./data/images
python run_cv.py --metadata ./data/metadata.csv --img_dir ./data/images
python scripts/plot_inkjet_results.py
```

Single-run reproduction can also be performed with `python train.py -metadata ./data/metadata.csv -img_dir ./data/images -sep_loss_weight 0.01 -eval_after` followed by `python evaluate.py -checkpoint results/train/cdm_best.pt -metadata ./data/metadata.csv -img_dir ./data/images -num_trials 100`.

5.5. Inkjet Print Quality Control Experiments

5.5.1. Dataset

The public FTI_Zer0P inkjet print quality dataset (Profactor, 2024) contains 1,204 inkjet print images captured from 15 building components, organised into 53 print datasets with up to three print areas per component. Of these images, 960 are labelled as GOOD or BAD for the relevant print-quality features.

Feature extraction. YOLOv8n detects and crops features from template images; each crop is labelled with template type, feature type, quality label (GOOD/BAD), and bounding box coordinates.

Sample Predictions - Correctly Classified Examples

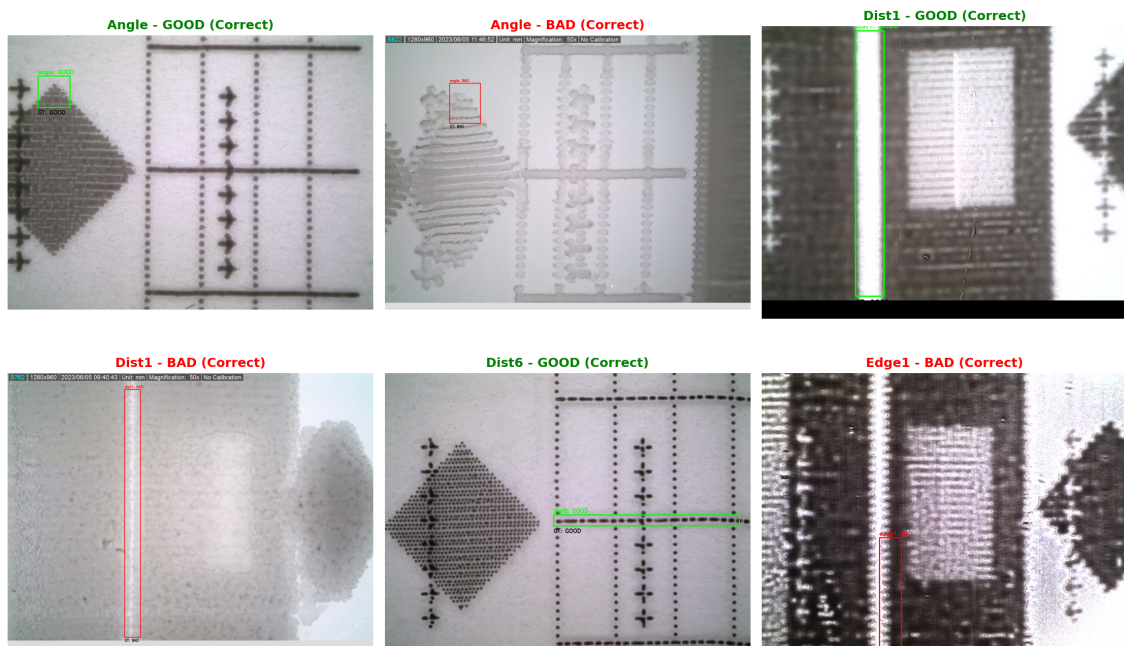


Figure 5.1.: Example images from the inkjet print quality dataset with YOLO detections and CDM predictions. Green bounding boxes indicate GOOD quality predictions; red boxes indicate BAD quality predictions. Feature types shown: angle, dist1, dist6, edge.

Per-Feature Class Balance (GOOD : BAD ratio):

- dist1: 1.82 : 1
- dist6: 1.94 : 1
- edge1–edge4: 1.68 : 1–2.31 : 1

Table 5.2.: Inkjet print quality dataset statistics. Public dataset counts follow the Zenodo record; retained CDM evaluation counts describe the processed YOLO feature-crop representation.

Property	Value
Public images	1,204
Public labelled images	960
Building components	15
Print datasets	53
Processed source-image groups	573
Resolution	1920 × 1080 pixels
Template types	3 (A, B, C)
Feature types	8 (dist1, dist6, edge1–edge4, angle, dots)
Quality labels	2 (GOOD, BAD)
Retained feature-level crops	6,408
GOOD : BAD ratio	~2:1 overall (varies by feature)

- angle: 9.67 : 1 (most imbalanced)
- dots: 3.45 : 1

5.5.2. Methods Under Comparison

The retained public experiment is the **YOLO + CDM** pipeline: YOLOv8n detects feature regions, and a multi-head CDM classifies each crop via differential reconstruction error under GOOD and BAD conditions. Only results regenerable from Inkjet00D, the public FTI_Zer0P dataset, and the released CDM weights are reported.

5.5.3. Evaluation Protocol

Cross-validation. Stratified 5-fold cross-validation with image-level splitting across 573 source-image groups (random seed 42, fixed across all λ settings so the same fold partitions are evaluated for every condition); all crops from the same source image stay in the same fold, preventing leakage. Separation-loss ablation trains the CDM with $\lambda \in \{0.0, 0.01, 0.02, 0.05\}$ to evaluate whether the separation loss improves quality classification.

Metrics. Primary: AUROC (mean $\pm$ std across 5 folds); secondary: Accuracy, F1, Precision, Recall. Paired t -tests and Wilcoxon signed-rank tests with Holm correction are reported for λ differences. All retained evaluations share the same image-level splits, YOLO detections, and evaluation codepath.

6. Results and Analysis

This chapter presents the experimental results for both tracks: CIFAR-10 OOD detection (Sections 6.1–6.5) and inkjet quality classification (Section 6.6). For $n \geq 5$ (inkjet folds), we report paired t-test and Wilcoxon results with Holm correction; for $n < 5$ (CIFAR seed-level), we report only descriptive comparisons.

Seed protocol. For the CIFAR-10 track, the two retained multi-seed separation loss weights ($\lambda = 0.01$, used for several design ablations, and $\lambda = 0.02$, the final selected weight) were each re-trained over three independent seeds (42, 123, and 456). The no-separation baseline ($\lambda = 0.0$) was later re-evaluated over the same three seeds in the final consistency pass. The seed-stability figure in this chapter refers to the final selected $\lambda = 0.02$ configuration. The training-summary figure retains the earlier three-seed $\lambda = 0.01$ checkpoint because that is the complete reproducible checkpoint series preserved in the submission artefacts. The K-ablation runs and the scoring-strategy comparison reuse the single seed-42 checkpoint at $\lambda = 0.01$ for compute-budget reasons, so the variance bars shown in those ablations should be interpreted as descriptive rather than as proper seed-level uncertainty estimates. All other λ values ($\{0.001, 0.05, 0.1\}$) are evaluated with seed 42 only. Because $n < 5$ even for the three-seed configurations, we do not report hypothesis tests on CIFAR-10 and instead follow the descriptive reporting convention stated above.

6.1. CIFAR-10 Binary CDM: Main Results

6.1.1. Within-Distribution Split

Figure 6.1 summarises AUROC values for checkpoint-level validation across three seeds. Table 6.1 reports the reproducible core metrics used for thesis claims.

Within-CIFAR Performance (core reproducible). Using current raw-score tensors (seed 42, $\lambda = 0.02$, $K = 100$), the binary CDM achieves 98.98% AUROC with 4.7% FPR95 on the airplane-vs-rest split; Figure 6.2 summarises the retained three-seed $\lambda = 0.01$ checkpoint reused for K , timestep, and scoring ablations.

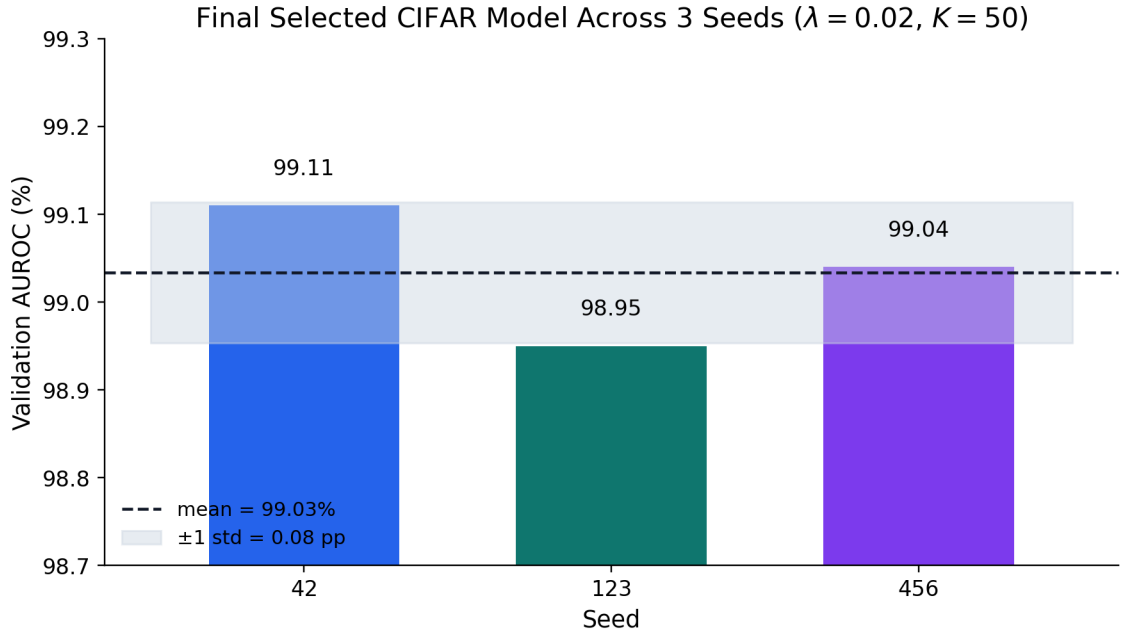


Figure 6.1.: Validation AUROC across training seeds (42/123/456) from checkpoint metadata. This figure is used as a stability summary; core quantitative claims are taken from reproducible raw-score evaluation (Table 6.1).

Table 6.1.: Binary CDM OOD detection performance from reproducible artefacts. Core claims use only metrics reproducible from current raw score files (seed 42, $\lambda=0.02$, $K=100$, difference scoring). $K=100$ is used here for statistical stability; ablation studies use $K=50$ (Section 5.4).

Dataset	Type	AUROC (%) $\uparrow$	FPR95 (%) $\downarrow$	AUPRC (%) $\uparrow$
Core reproducible metrics (current raw scores)				
CIFAR-10 (airplane vs others)	Within	98.98	4.7	99.87
CIFAR-100	Near OOD	96.97	14.8	99.65
Places365	Far OOD	96.50	15.4	99.57
FashionMNIST	Far OOD	94.03	20.5	99.16
Textures (DTD)	Far OOD	92.84	30.1	95.97
SVHN	Far OOD	90.50	27.0	99.38

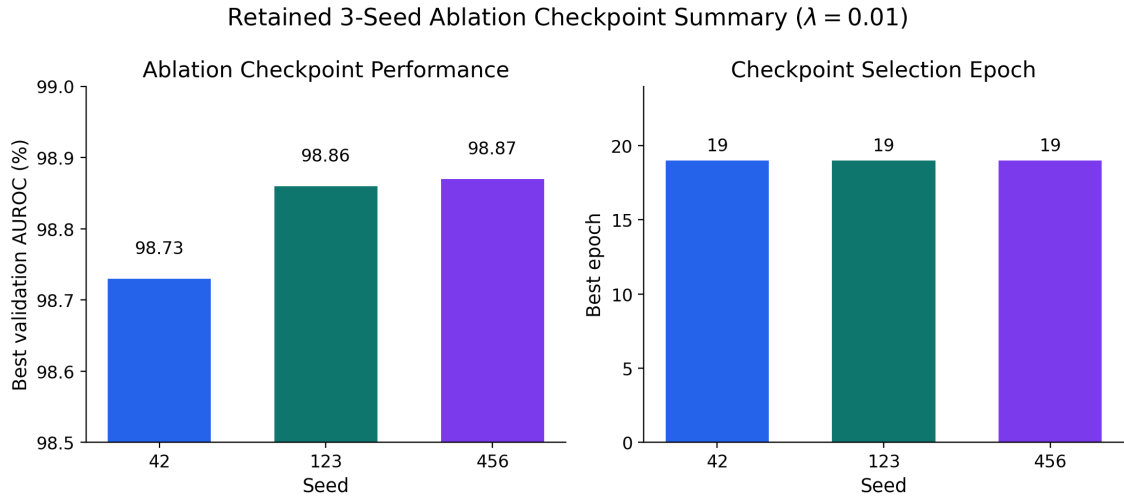


Figure 6.2.: Training summary from checkpoint metadata: best validation AUROC and best epoch for seeds 42, 123, and 456 for the retained $\lambda=0.01$ ablation checkpoint.

6.1.2. External OOD Generalisation

Under the public-evidence protocol, core external claims use only datasets with released raw-score tensors. This yields five reproducible external benchmarks: CIFAR-100, Places365, FashionMNIST, Textures, and SVHN.

Reproducible external profile. AUROC ranges from 90.50% (SVHN) to 96.97% (CIFAR-100), with a mean of 94.17% across the five reproducible external sets, each evaluated against the full CIFAR-10 test reference pool described in Section 5.4.3. Performance is highest for CIFAR-100 (96.97%) and Places365 (96.50%) and lowest for SVHN (90.50%) and Textures (92.84%); this ordering does not follow a simple near-vs-far-OOD gradient and is discussed in Section 7.1.

6.2. Contextualisation Against One-Class Baselines

Table 6.2 compares our binary CDM against published one-class novelty detection methods on the CIFAR-10 airplane class (one-vs-rest protocol).

Without the separation loss, the binary CDM is competitive on average but highly seed-sensitive. With $\lambda=0.02$, performance becomes both stronger and substantially more stable, exceeding the listed one-class baselines in this single-class airplane-vs-rest setting *under this asymmetric protocol*: our CDM trains with an OOD-proxy class while the one-class baselines train on ID data only, so the margin should be read as contextual rather than as a like-for-like ranking (see caveats below).

Table 6.2.: Comparison with one-class novelty detection baselines on the CIFAR-10 airplane class (one-vs-rest). Published values are taken from the original papers or as reproduced by (Reiss, Cohen, et al., 2021). Our CDM rows use $K=50$ and three-seed means over seeds 42/123/456 where available. The $\lambda=0$ row reports the no-separation-loss baseline; the $\lambda=0.02$ row reports the best-tested separation-loss setting. This comparison is asymmetric because the CDM is trained with an OOD-proxy class, whereas the one-class baselines are trained only on ID samples.

Method	Type	AUROC (%)	Source
OC-SVM (raw pixels)	One-class	63.0	(Ruff et al., 2018)
Deep SVDD	One-class	61.7	(Ruff et al., 2018)
DROCC	One-class	81.7	(Goyal et al., 2020)
CSI	Contrastive	89.8	(Tack et al., 2020)
PANDA	Pretrained + OC	95.4	(Reiss, Cohen, et al., 2021)
Mean-Shifted C.L.	Contrastive	97.5	(Reiss and Hoshen, 2023)
Binary CDM ($\lambda=0$)	Generative	92.52 ± 11.07	Ours
Binary CDM ($\lambda=0.02$)	Generative + sep. loss	99.03 ± 0.07	Ours

Caveats. These comparisons should be interpreted with care: (i) published values use different backbones, training budgets, and hyperparameter selection procedures; (ii) our method trains on both ID and OOD-proxy data, while pure one-class methods see only ID samples; (iii) we report results for a single ID class (airplane), and per-class variance across all 10 CIFAR-10 classes can be substantial. A comprehensive multi-class evaluation is identified as future work in Section 7.6. For broader context, the recent task-oriented survey by Lu et al. (2025) catalogues state-of-the-art results across methods and benchmarks.

6.3. Ablation Studies

6.3.1. Number of Monte Carlo Trials (K)

Table 6.3 and Figure 6.3 show how OOD performance varies with K (seed-42 checkpoint, $\lambda = 0.01$).

Note on checkpoint. This ablation uses the seed-42 checkpoint at $\lambda = 0.01$ (not the best-tested $\lambda = 0.02$ setting) for compute-budget reasons. The K -AUROC curve shape is expected to be similar at $\lambda = 0.02$ (the separation loss does not change the structure of the MC averaging), but exact numbers at the best-tested weight have not been independently verified; this caveat is noted in the RQ3 answer (Section 8.1.1).

Table 6.3.: Effect of number of Monte Carlo trials K on OOD scoring. Evaluated on CIFAR-10 binary test set using the seed-42 $\lambda=0.01$ checkpoint. Diminishing returns beyond $K = 25$.

K (trials)	AUROC (%) $\uparrow$	FPR@95%TPR $\downarrow$	Time (s)
1	91.00	40.8	97.9
5	97.24	14.3	486.3
10	98.19	9.4	972.9
25	98.52	7.3	2 431.8
50	98.64	6.6	4 861.1
100	98.69	6.6	9 723.6

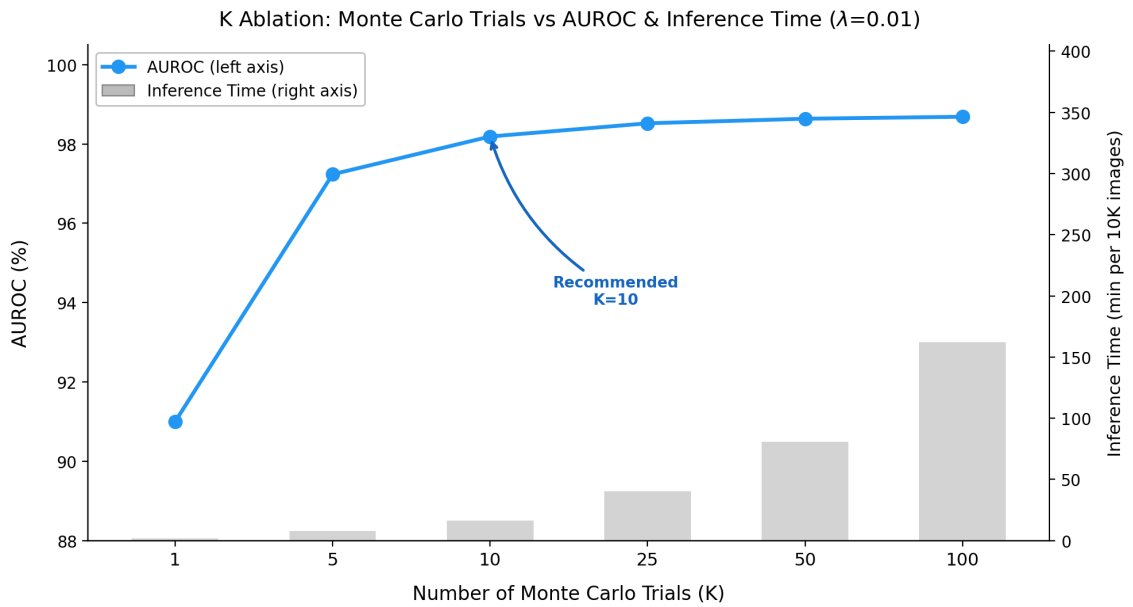


Figure 6.3.: Effect of K on AUROC (left axis) and inference time for 10K images (right axis), evaluated with the seed-42 $\lambda=0.01$ checkpoint. The curve flattens after $K = 25$; $K = 10$ offers the best accuracy–efficiency trade-off at $5\times$ speedup over $K = 50$.

Single-shot baseline. Even $K = 1$ (one random timestep per image) achieves 91.0% AUROC in under 2 minutes per 10K images, demonstrating that a single denoising step carries meaningful OOD signal.

Rapid saturation. Performance improves sharply from $K = 1$ to $K = 10$ (+7.2% AUROC), then diminishes: $K = 25$ yields only +0.3% over $K = 10$, and $K = 100$ only +0.05% over $K = 50$. This confirms that variance reduction from averaging timestep samples saturates quickly.

Practical recommendations. $K = 10$ achieves 98.2% AUROC at $5\times$ speedup over $K = 50$ (16.2 vs. 81 minutes per 10K images) and is recommended for throughput-sensitive applications. $K = 50$ is used as the default throughout this thesis for maximum reproducibility.

6.3.2. Timestep Sampling Strategy

Table 6.4.: Comparison of timestep sampling strategies using the seed-42 $\lambda=0.01$ checkpoint and $K=50$.

Strategy	CIFAR AUROC (%)	SVHN AUROC (%)
Uniform	98.9	95.4
Mid Focus	98.5	93.8
Stratified	98.8	95.0

Uniform sampling is the best-tested strategy. Sampling $t \sim \mathcal{U}[1, 1000]$ achieves the highest AUROC on both datasets (Within-CIFAR: 98.9%, SVHN: 95.4%). This indicates that OOD-discriminative information is distributed across the full range of noise levels.

Stratified is equivalent. Dividing $[1, 1000]$ into equal-width bins and sampling one timestep per bin achieves 98.8%/95.0%, essentially identical to uniform ($< 0.1\%$ gap).

Mid-focus underperforms. Concentrating samples at intermediate timesteps ($t \sim \mathcal{N}_{\text{trunc}}(\mu=300, \sigma=150)$) reduces performance to 98.5%/93.8%. The per-timestep analysis (Figure 6.7) shows that excluding the tails of the timestep range discards useful OOD signal.

6.3.3. OOD Scoring Method

Contrastive scoring is essential. Using only the ID-class reconstruction error (e_{ID}) without contrasting against the OOD-class error achieves only 78.3% Within-CIFAR AUROC and collapses to 20.2% on SVHN, essentially random. The binary conditioning structure ($c=0$ for airplane, $c=1$ for rest) enables the comparison, but the ablation specifically shows that the OOD score must contrast reconstruction under each condition rather than rely on magnitude alone.

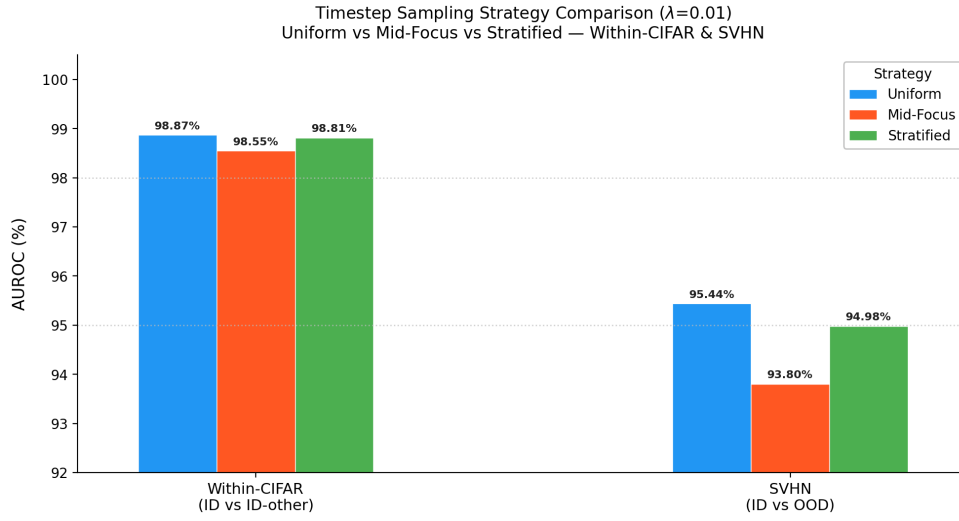


Figure 6.4.: AUROC for three timestep sampling strategies on the within-CIFAR binary split and SVHN ($K = 50$, seed-42, $\lambda=0.01$). Uniform sampling is best; stratified is equivalent; mid-focus underperforms on both datasets.

Table 6.5.: Comparison of OOD scoring methods (seed-42 $\lambda=0.01$ checkpoint, $K=50$ trials). Difference and ratio perform similarly within CIFAR-10; ratio marginally better on external SVHN OOD. ID-error (ID-only, non-contrastive scoring) is much worse, confirming that the two conditional errors must be contrasted.

Scoring Method	CIFAR AUROC (%) $\uparrow$	CIFAR FPR95 (%) $\downarrow$	SVHN AUROC (%) $\uparrow$
difference	98.69	6.3	94.1
ratio	98.62	6.6	96.1
id_error	78.30	67.0	20.2

Difference vs. ratio. The absolute difference $e_{ID} - e_{OOD}$ (i.e. $e_0 - e_1$) performs marginally better Within-CIFAR (98.69% vs. 98.62%), while the ratio e_{ID}/e_{OOD} performs better on SVHN (96.1% vs. 94.1%). We adopt the difference formulation as default throughout this thesis due to its lower FPR95 Within-CIFAR (6.3% vs. 6.6%) and simpler interpretability.

6.4. Separation Loss Analysis

The separation loss is the primary novel contribution of this thesis. It adds an explicit training signal that pushes apart the class-conditional noise predictions, indirectly widening the reconstruction-error gap between ID and OOD-conditioned predictions. We present a complete λ sweep over six values followed by cross-domain analysis on

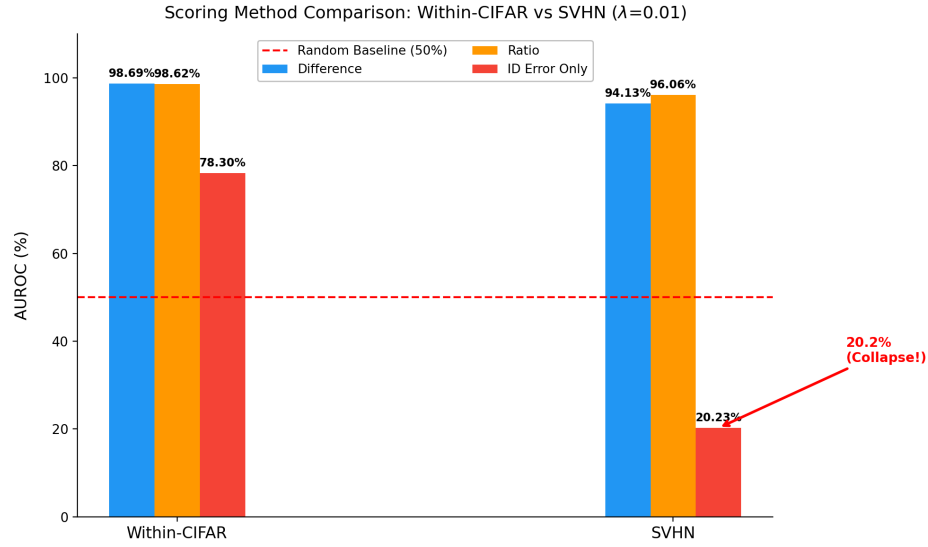


Figure 6.5.: AUROC comparison for three scoring methods on Within-CIFAR and SVHN ($K = 50$, seed-42, $\lambda=0.01$). Difference and ratio scoring both perform well; ID-error-only scoring collapses on SVHN (20.2%), demonstrating the necessity of contrastive scoring.

the inkjet dataset.

6.4.1. Within-CIFAR Lambda Sweep

Table 6.6.: Effect of separation loss weight λ on OOD detection performance. Within-CIFAR AUROC uses three-seed mean $\pm$ std for $\lambda \in \{0.0, 0.01, 0.02\}$ and seed-42 for the remaining λ values. SVHN AUROC uses seed-42 evaluation. [†] marks documented artefact points kept for traceability.

λ	Within-CIFAR AUROC (%)	Best Epoch (seed 42)	SVHN AUROC (%)
0.0	92.52 ± 11.07	79	100.0 [†]
0.001	97.32	19	92.0
0.01	98.82 ± 0.06	19	90.5
0.02	99.03 ± 0.07	29	96.6
0.05	98.51	19	97.3
0.1	96.67	149	86.9 [†]

No separation loss ($\lambda = 0.0$). Without an explicit separation signal, the three-seed mean is $92.52\% \pm 11.07\%$ AUROC (individual: 79.73%, 98.81%, 99.01%). Conditioning alone is

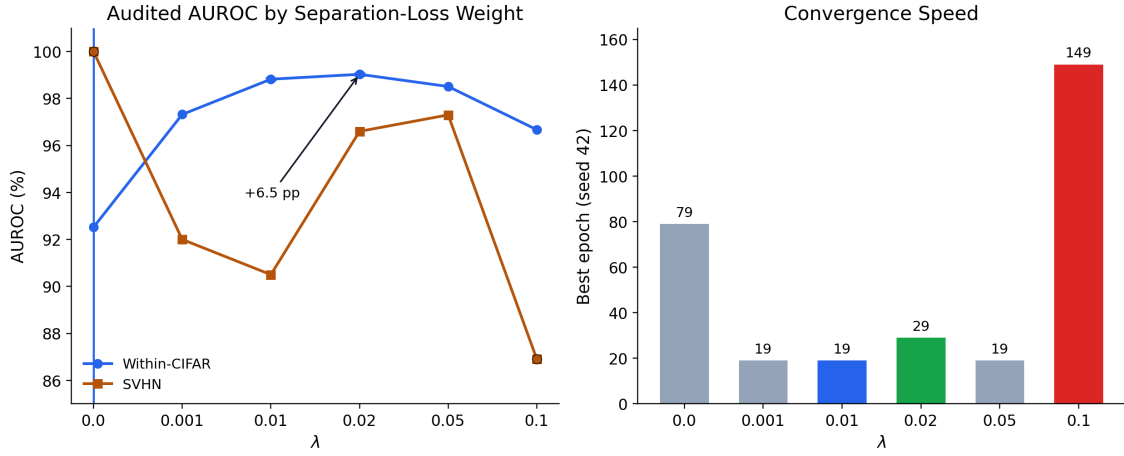


Figure 6.6.: Effect of separation loss weight λ on Within-CIFAR and SVHN AUROC (left) together with the seed-42 best epoch for each retained checkpoint (right), generated from the final consistency ablation summary. Crosses in the SVHN series mark documented artefact points.

therefore not uniformly weak, but highly seed-sensitive: one seed reproduces the earlier 80% regime while two seeds approach the separated model.

Large jump at small λ . The seed-42 $\lambda = 0.001$ run reaches 97.32%, showing that even a weak separation signal can preserve strong performance without the large seed sensitivity seen at $\lambda = 0$.

Best-tested at $\lambda = 0.02$ (three-seed). Performance peaks at $\lambda = 0.02$: $99.03\% \pm 0.07\%$ (individual: 99.11%, 98.95%, 99.04%), a total +6.5 pp gain over the three-seed $\lambda = 0$ baseline and the best within-CIFAR result in this study. The more striking difference is the much tighter seed spread at $\lambda = 0.02$.

Overfitting at $\lambda = 0.1$. High weights degrade performance (96.67%, unusually late convergence at epoch 149 vs. the typical 15–25) as the separation objective dominates the denoising loss.

SVHN artefacts. The $\lambda = 0.0$ SVHN value (100%) is a known scoring-direction artefact from degenerate near-zero difference scores. Excluding $\lambda \in \{0.0, 0.1\}$, SVHN AUROC increases consistently up to $\lambda = 0.05$ (97.3%), then drops at $\lambda = 0.1$ (86.9%).

6.4.2. Effect on Score Distributions

Inspection of the reproducible raw-score tensors shows a consistent pattern: ID scores are centred near zero or slightly negative values, while OOD scores shift to higher positive values across the retained external datasets. The amount of overlap tracks the observed AUROC difficulty; SVHN and Textures exhibit more overlap than CIFAR-100, matching their lower AUROC values in Table 6.1. The retained figure set is regenerated only from the public raw-score artefacts listed in the reproducibility appendix.

6.5. Qualitative Analysis

6.5.1. Per-Timestep Error Analysis

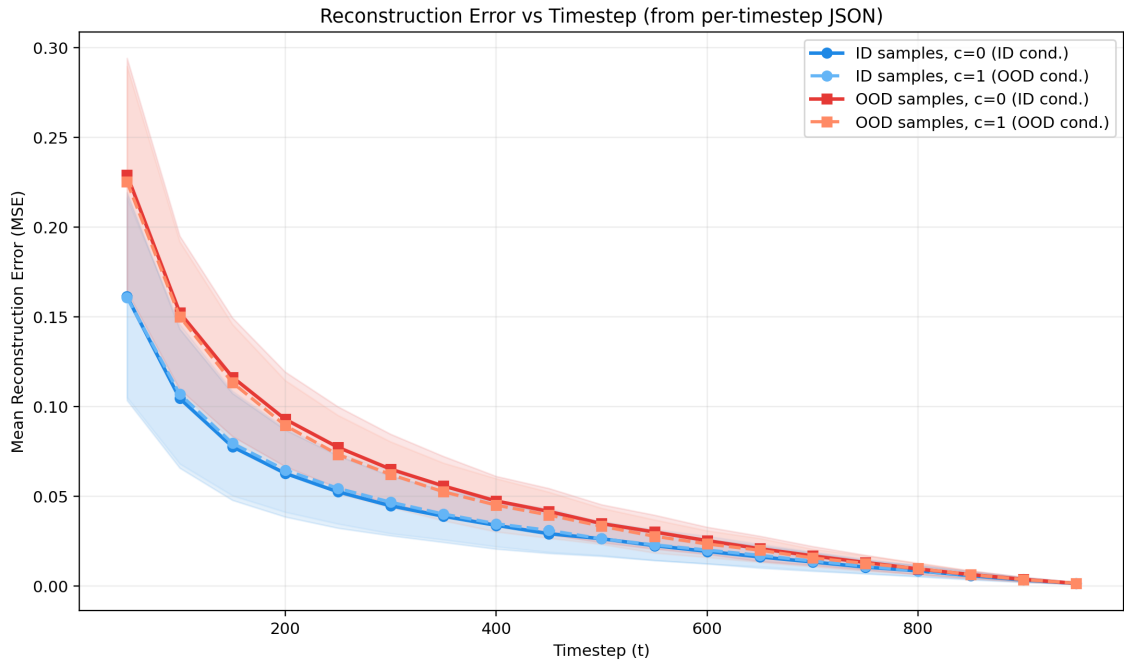


Figure 6.7.: Mean reconstruction error as a function of timestep t from the reproducible `per_timestep` JSON block. Curves are shown for ID and OOD samples under both conditions ($c = 0$ and $c = 1$), with one-standard-deviation bands.

The ID–OOD error gap peaks at intermediate noise levels and shrinks at the extremes: near-clean inputs ($t < 50$) and near-pure noise ($t > 900$) yield similar errors for both ID and OOD, while the informative mid-range provides the OOD-discriminative signal captured by full-range uniform sampling.

Calibration and decision boundary. Figure 6.8 shows the within-CIFAR OOD score distribution for ID and OOD samples at the operating threshold calibrated to $\text{TPR} = 95\%$. The separation between the two score distributions is clear: ID samples (airplane) cluster around negative or near-zero scores (i.e., better reconstruction under the ID condition), while OOD samples accumulate at higher positive values, consistent with the per-timestep error analysis in Figure 6.7, which shows that OOD samples incur higher reconstruction error under both class conditions. The figure confirms that the $s(x) > \tau$ threshold at $\text{TPR} = 95\%$ corresponds to the FPR_{95} values reported in Table 6.1.

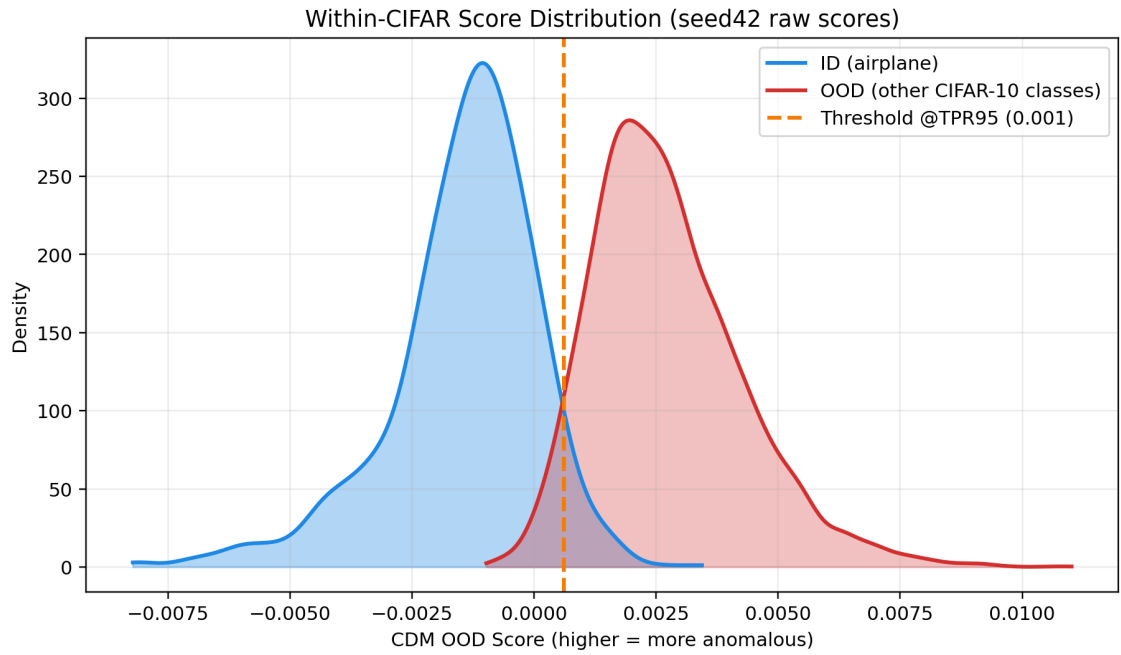


Figure 6.8.: Within-CIFAR score distribution with the operating threshold selected at $\text{TPR} = 95\%$. At this threshold the model achieves 95% TPR, appropriate for anomaly detection applications.

6.6. Inkjet Print Quality Classification Results

We evaluate the public YOLO+CDM quality-classification pipeline on the public FTI_Zer0P inkjet dataset under 5-fold stratified CV. Section 6.6.1 summarises the retained public CDM evidence; Section 6.6.2 analyses per-feature CDM performance; Section 6.6.3 reports the separation-loss ablation.

6.6.1. Public YOLO+CDM Summary

The retained public inkjet evidence is the YOLO+CDM pipeline released in Inkjet00D. At $\lambda = 0$ and $K = 100$, the pipeline reaches 0.8673 ± 0.0230 AUROC under 5-fold image-level cross-validation (Table 6.8). This result is reproducible from the public FTL_Zer0P dataset, the released code, and the published inkjet CDM weights. Earlier internal supervised and anomaly-detection comparisons are excluded from retained thesis claims because their full code, configurations, splits, and result artefacts are outside the released public package.

6.6.2. Per-Feature Analysis

The CDM’s per-feature results explain where the crop-based representation remains sufficient and where local quality cues become limiting.

Table 6.7.: Per-feature AUROC on the Inkjet QC dataset (5-fold CV, mean $\pm$ std). [†] angle has fewer than 5 BAD samples per fold; AUROC is unreliable.

Feature	$\lambda = 0.0$	$\lambda = 0.01$	$\lambda = 0.02$	$\lambda = 0.05$
Angle [†]	0.8166 ± 0.1381	0.7727 ± 0.1580	0.7682 ± 0.1836	0.7378 ± 0.1593
Distance 1	0.8866 ± 0.0729	0.8508 ± 0.0768	0.8274 ± 0.1457	0.8828 ± 0.0581
Distance 6	0.9362 ± 0.0666	0.9423 ± 0.0737	0.9393 ± 0.0750	0.9473 ± 0.0710
Dots	0.9557 ± 0.0353	0.9655 ± 0.0247	0.9474 ± 0.0400	0.9413 ± 0.0394
Edge 1	0.7958 ± 0.1376	0.8410 ± 0.1379	0.7823 ± 0.1607	0.8574 ± 0.1283
Edge 2	0.8129 ± 0.0307	0.8426 ± 0.0443	0.7993 ± 0.0602	0.8402 ± 0.0283
Edge 3	0.7441 ± 0.0757	0.7008 ± 0.0390	0.7774 ± 0.0647	0.7349 ± 0.0596
Edge 4	0.7617 ± 0.0985	0.7529 ± 0.0543	0.7643 ± 0.0898	0.7775 ± 0.1125

A consistent feature-difficulty gradient appears across all λ settings. Distance features (dist6: 0.937–0.947, dist1: 0.851–0.887 for CDM) reflect clear geometric differences between GOOD and BAD prints that are visible even in a localised crop. Edge roughness features (edge2–edge4: 0.741–0.843) require fine-grained texture analysis and show higher cross-fold variance.

6.6.3. Separation Loss on Inkjet Data

In contrast to the CIFAR-10 results, the separation loss does not improve performance on the inkjet dataset. All tested weights ($\lambda \in \{0.01, 0.02, 0.05\}$) produce AUROC values within ± 0.016 of the $\lambda = 0$ baseline (0.867), smaller than the cross-fold standard deviation (± 0.023). Using 5-fold paired comparisons against $\lambda = 0$, the raw paired t-test p -values are 0.371 ($\lambda = 0.01$), 0.143 ($\lambda = 0.02$), and 0.967 ($\lambda = 0.05$). After Holm

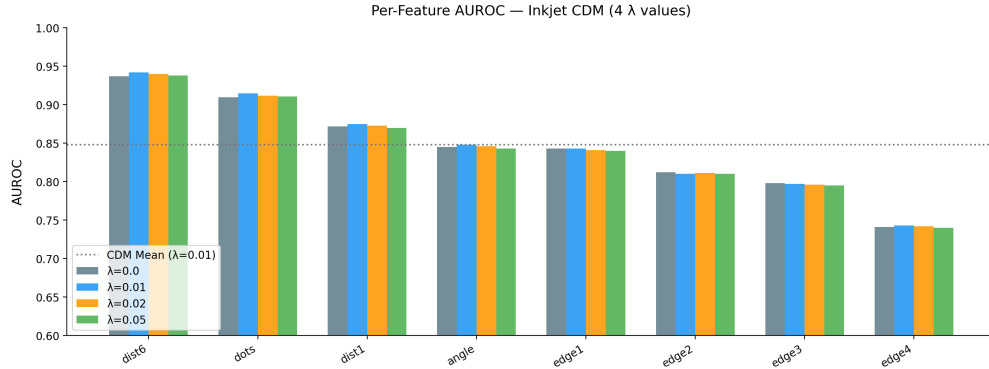


Figure 6.9.: Per-feature AUROC for the CDM across four λ values (5-fold CV). Distance features (dist6, dots) are easiest; edge roughness features (edge3, edge4) are hardest for all settings.

Table 6.8.: Separation loss ablation on Inkjet QC (5-fold stratified CV, $K = 100$). Results are mean $\pm$ std across folds. No λ value is significant against baseline after Holm correction (paired t-test and Wilcoxon).

λ	AUROC $\uparrow$	Accuracy $\uparrow$	FPR@95%TPR $\downarrow$
0.0 (baseline)	0.8673 ± 0.0230	0.8094 ± 0.0151	0.5631 ± 0.1697
0.01	0.8628 ± 0.0286	0.7928 ± 0.0291	0.5516 ± 0.1841
0.02	0.8510 ± 0.0326	0.8003 ± 0.0246	0.6240 ± 0.1334
0.05	0.8670 ± 0.0256	0.8071 ± 0.0241	0.5700 ± 0.1948

correction, all adjusted p -values remain above 0.05 (0.742, 0.429, 0.967), confirming no significant improvement.

Remark on single-split vs. cross-validation. A single-seed 80/20 split (seed 42, $N = 266$ test samples) does show $\lambda = 0.01$ ahead of the baseline: 0.860 vs. 0.833 (+2.8 pp), with a larger FPR@95 improvement (0.626 vs. 0.816). This apparent gain disappears under 5-fold CV (0.863 vs. 0.867, difference = 0.004), indicating that the single-split result is within the variance of a single random fold rather than a reliable improvement. With only 573 processed source-image groups (6,408 feature crops), the cross-fold standard deviation (± 0.023) exceeds the observed single-split gain, so 5-fold CV lacks the statistical power to confirm a difference of 2.8 pp; the cross-validation estimate is the more trustworthy summary.

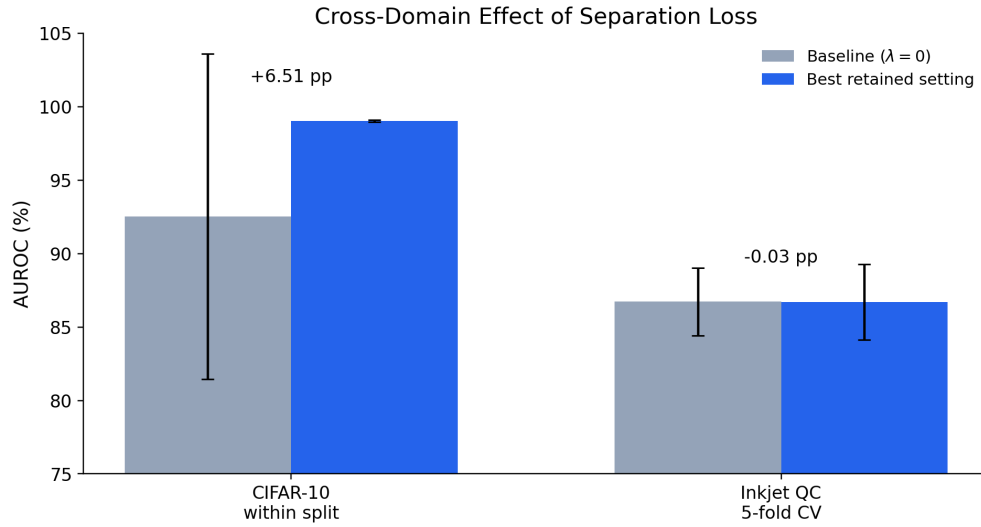


Figure 6.10.: Cross-domain effect of separation loss on CDM AUROC. The CIFAR-10 panel compares the three-seed $\lambda = 0$ baseline with the best-tested $\lambda = 0.02$ setting; the inkjet panel shows 5-fold CV means across λ values, all within the baseline fold variability.

6.6.4. Cross-Domain Analysis

Table 6.9.: Cross-domain comparison of separation loss effect. CIFAR-10 uses three-seed Within-CIFAR AUROC from the final consistency separation-loss summary; Inkjet uses 5-fold CV mean $\pm$ std.

Dataset	Baseline ($\lambda = 0$)	Best sep. loss	Δ AUROC
CIFAR-10	0.925 ± 0.111	0.9903 ± 0.0007 ($\lambda = 0.02$)	+6.5 pp
Inkjet QC	0.8673 ± 0.0230	0.8670 ± 0.0256 ($\lambda = 0.05$)	-0.03 pp

Inkjet significance: paired t-test + Wilcoxon with Holm correction, all adjusted $p > 0.05$.

The domain-dependent result reveals the boundary conditions of the separation loss relative to Section 3.5.1. Three factors likely explain the contrast with CIFAR-10: (i) the retained public inkjet pipeline uses only 573 processed source-image groups (6,408 feature crops) versus CIFAR-10's 50,000 images, making the separation gradient noisier; (ii) GOOD and BAD inkjet prints differ in fine defect details rather than a large semantic shift, demanding higher reconstruction precision; and (iii) the CDM is trained from scratch on a narrow domain with limited diversity. Thus, the separation loss is strong for the evaluated CIFAR-10 binary OOD setting but requires validation on small domain-specific datasets before deployment.

7. Discussion

This chapter interprets the experimental results presented in Chapter 6. We discuss the key findings for each track separately, draw cross-domain insights from comparing the binary CIFAR-10 and inkjet quality control settings, and examine the practical and theoretical implications of the results. Known limitations and directions for future work conclude the chapter.

7.1. Key Findings: CIFAR-10 Binary CDM

7.1.1. The Separation Loss Dominates the CIFAR-10 Ablation

The separation loss mainly acts as a stabiliser. The three-seed $\lambda = 0$ baseline reaches $92.52\% \pm 11.07\%$ AUROC, but that mean hides extreme seed sensitivity: the individual runs span 79.73%, 98.81%, and 99.01%. Adding the separation loss moves the mean to $99.03\% \pm 0.07\%$ at $\lambda = 0.02$, a smaller +6.5pp gain than the seed-42-only comparison suggested, but with a dramatic reduction in variance. The effect remains non-linear: $\lambda = 0.001$ already yields 97.32%, the best-tested setting is $\lambda = 0.02$, and degrades at $\lambda = 0.1$ (96.67%) as the separation objective dominates the denoising loss.

7.1.2. Contrastive Scoring Is Essential

The OOD score must contrast reconstruction errors under both binary conditions. Using only the ID-class error collapses to 20.2% AUROC on SVHN (essentially random) because global shifts in reconstruction scale confound the absolute magnitude. The contrastive score $s(x) = e_0(x) - e_1(x)$ cancels these confounders, measuring only the model’s relative preference for one condition over the other. Thus, the ablation supports the need for contrastive use of the two conditions; it does not by itself isolate an architecture with conditioning removed.

7.1.3. External OOD Generalisation Reveals Learnt ID Manifold

Across five reproducible external benchmarks, AUROC ranges from 90.50% to 96.97%, with each benchmark evaluated against the full CIFAR-10 test reference pool. Although

CIFAR-100 is a near-OOD dataset (sharing low-level visual statistics with CIFAR-10), it achieves the highest AUROC (96.97%), higher than far-OOD datasets such as Textures (92.84%) and SVHN (90.50%). As noted in Section 3.2.3, this counterintuitive result arises because the airplane class is under-represented in CIFAR-100, creating a strong distributional gap that the CDM exploits effectively. The result demonstrates that semantic proximity to the source dataset does not deterministically predict detection difficulty; what matters is proximity to the specific ID class manifold (airplane). SVHN and Textures are harder despite being far-OOD because their low-level statistics (digit strokes, repeating textures) overlap with certain airplane image regions.

7.1.4. Ablation Insights: Efficiency and Design Choices

Monte Carlo trials. Performance improves rapidly to $K=10$ then saturates (Section 6.3), making $K=10$ the practical recommendation. The K -AUROC curve was benchmarked at $\lambda=0.01$; transferability of this saturation point to the $\lambda=0.02$ regime was not independently verified. Even $K=1$ provides useful signal, suggesting that a single denoising step is a non-trivial distributional indicator.

Timestep sampling. Uniform sampling across $[1, T]$ is the best-tested strategy. The per-timestep analysis (Figure 6.7) shows OOD-discriminative information distributed across the full noise range; concentrating on intermediate timesteps discards useful signal from the tails.

Scoring method. Difference and ratio scoring perform comparably on the within-CIFAR split, with ratio generalising slightly better to external datasets. We retain difference scoring as the default for its lower FPR95 and simpler interpretation.

7.2. Key Findings: Inkjet Print Quality Control

7.2.1. Public YOLO+CDM Performance

The retained public inkjet evidence shows that the YOLO+CDM pipeline is a viable quality-classification signal on FTI_Zer0P, reaching 0.8673 ± 0.0230 AUROC at $\lambda = 0$ under 5-fold image-level cross-validation. This is an encouraging result for a crop-based generative classifier, but it should be interpreted within the released evidence boundary: the thesis no longer uses unreleased multi-method baseline comparisons as support for ranking claims.

7.2.2. Feature Difficulty Is Heterogeneous

Per-feature AUROC varies substantially across the public CDM folds (Table 6.7, Figure 6.9). Distance and dot features are consistently easier, while edge roughness features are more variable. This suggests that the local crop signal is sufficient for some geometric defects but weaker for subtle texture or boundary defects.

7.2.3. Separation Loss Does Not Transfer Automatically

Unlike the CIFAR-10 track, non-zero separation-loss weights do not produce a statistically significant gain on the public inkjet folds (Table 6.8). The negative result is useful: the loss can sharpen class-conditional reconstruction boundaries when data scale and class separation are favourable, but it is not a universal recipe for small, fine-grained industrial datasets.

7.2.4. YOLO Detection as a Pipeline Dependency

The two-stage pipeline introduces a detection dependency that is not independently evaluated: we do not compare CDM classifications between YOLO-detected and ground-truth crops. The YOLO model achieves 95.04% mAP@0.5, suggesting detection errors are infrequent but not negligible. An ablation isolating the detection bottleneck is recommended for future work.

7.3. Cross-Domain Insights

7.3.1. The Separation Loss Has Boundary Conditions

The most important cross-domain finding is that the separation loss is highly effective on CIFAR-10 but not significant on inkjet data (Table 6.9). The technique appears to depend on sufficient data scale and a clear visual gap between conditions; in the public inkjet setting, the available 1,204 images and subtle GOOD/BAD differences make the separation signal less reliable than in CIFAR-10. This negative result bounds the applicability of the technique and prevents over-generalisation.

7.3.2. Contrastive Scoring Transfers Across the Two Tracks

Both evaluated tracks support the use of scores that compare reconstruction errors under two conditions. Contrastive scoring normalises away global reconstruction scale differences and focuses on the model's relative preference for one condition over the other, which is the key insight separating conditional diffusion detectors from absolute reconstruction-error methods.

7.3.3. Public Evidence Boundaries Matter

Both tracks reinforce a practical lesson about evidence quality. The CIFAR-10 track supports broad ablation claims because the relevant weights, scripts, and raw-score artefacts are public. The inkjet track supports narrower claims: the public YOLO+CDM pipeline, per-feature behaviour, and separation-loss ablation. Claims about broader method rankings or representation choices require equally complete public artefacts before they can be treated as retained thesis evidence.

7.4. Implications

7.4.1. Theoretical Implications

Our results support conditional diffusion models as generative classifiers: scoring via reconstruction-error contrast can be interpreted as a likelihood-ratio-style comparison between competing conditions. The reproducible results are consistent with a support-based interpretation, although detection difficulty is not explained by a simple near-vs-far OOD ordering. The separation loss explicitly shapes the learnt manifold to maximise the inter-condition reconstruction gap, a useful principle in the large-scale CIFAR-10 setting studied here: when the training objective does not directly optimise the evaluation metric, a targeted auxiliary loss can improve the scoring signal.

7.4.2. Practical Implications

The binary CDM with separation loss ($\sim 68.8\text{M}$ parameters) achieves 99% AUROC at $K = 50$ and 98.2% at $K = 10$ (the K -AUROC curve was benchmarked at $\lambda=0.01$; saturation behaviour at $\lambda=0.02$ was not independently verified). It is recommended when high accuracy is required, inference cost is acceptable ($K = 10$: about 16.2 min per 10K images on an NVIDIA GeForce RTX 2080 Ti; $K = 50$: about 81.0 min), and the task has sufficient labelled data with a clear visual separation signal. Simpler discriminative alternatives are preferred for ultra-low latency or limited-data settings. In the inkjet comparison, the ResNet baselines use 224×224 inputs with ImageNet pre-training whereas the CDM operates on YOLO-cropped 128×128 feature images; absolute comparisons are therefore affected by both model family and input-resolution/context differences. For industrial QC specifically, the public inkjet results suggest that additional data scale, feature-specific analysis, and careful validation matter more than simply porting the CIFAR-10 separation-loss recipe. For quality inspection pipelines, the practical recommendation is to validate the public crop-based generative signal against the exact deployment constraints before drawing broader architecture or representation conclusions.

7.5. Limitations

7.5.1. Experimental Limitations

The evaluation is scoped to a single ID class (airplane), a single benchmark dataset (CIFAR-10), a single industrial application (inkjet QC), and external OOD datasets limited to natural images. Results should therefore be interpreted as strong evidence within these two settings rather than as universal claims. Extension to multi-class splits, higher-resolution datasets, diverse industrial domains, and adversarial/corrupted inputs would strengthen generalisability.

7.5.2. Methodological Limitations

Inference cost. Even at $K=10$, the binary CDM is $68.6\times$ slower than the measured discriminative ResNet-18 forward pass on the same RTX 2080 Ti, precluding real-time applications.

Training cost. Beyond inference, the separation-loss objective itself imposes a training-time overhead that is decoupled from the inference cost above. Each optimisation step requires three forward passes of the UNet: one for $\mathcal{L}_{\text{diffusion}}$ using the true label, plus two additional passes under the fixed $c=0$ and $c=1$ conditions to form $\mathcal{L}_{\text{sep}}$ (Equation 3.23), so training the binary CDM with $\lambda > 0$ costs approximately $3\times$ the per-step compute of the $\lambda = 0$ baseline and is correspondingly more expensive than a discriminative ResNet-18 fine-tuning run (which completes in $\sim 2\text{--}4$ hours on an A100). Practitioners evaluating the total computational investment of our method should account for both the training overhead and the inference overhead: the separation loss is cheap to add in code but not cheap to run, and this cost is separate from the K -dependent inference budget.

Binary framing. The binary setup (one ID vs. one proxy class) simplifies the general multi-class OOD problem; the proxy class is itself in-distribution data.

Separation loss tuning. No principled method exists for selecting λ ; it must be treated as a task-dependent hyperparameter selected via ablation.

Threshold calibration. All methods require threshold calibration assuming access to representative OOD or defective samples during validation.

7.5.3. Theoretical Limitations

Lack of formal guarantees. We provide intuitive explanations for why reconstruction error contrast indicates OOD status, but lack formal theoretical guarantees (e.g., PAC-style bounds). Informal Claim 2 in Section 3.2.3 provides heuristic support but not a rigorous proof.

No adversarial evaluation. We do not evaluate robustness to adversarial examples crafted to fool the OOD detector. Diffusion models have known vulnerabilities to adversarial perturbations that preserve low-level statistics while crossing decision boundaries.

7.6. Future Work

Methodological extensions. Multi-class binary CDM evaluation (all 10 CIFAR-10 classes as ID in turn); adaptive λ selection via meta-learning or validation-set AUROC; full-image or context-aware diffusion variants for inkjet, leveraging latent diffusion (Linhmans et al., 2024) or layer-wise reconstruction (Y. Yang et al., 2024); crop conditioning that incorporates spatial metadata.

Scaling and efficiency. Score distillation to a lightweight discriminative model (Wu et al., 2024); progressive coarse-to-fine evaluation ($K=1$ triage, $K=50$ for borderline cases); extension to higher resolutions ($256 \times 256+$) for medical and industrial settings.

Theoretical investigations. Formal bounds relating λ to AUROC improvement; adversarial robustness evaluation; application of reconstruction-error scoring to text, audio, and time-series modalities.

8. Conclusion

This thesis investigated conditional diffusion models as generative classifiers for two complementary tasks: out-of-distribution detection on CIFAR-10 benchmarks and industrial quality classification of inkjet experimental prints. On the CIFAR-10 within-binary (airplane-vs-rest) split, the final selected separation-loss setting is $\lambda=0.02$, which achieves $99.03\% \pm 0.07\%$ AUROC over three seeds at $K=50$; the reproducible seed-42 $K=100$ evaluation reaches 98.98% AUROC and 90.50–96.97% across five reproducible external datasets. On the public FTI_Zer0P inkjet dataset, the YOLO+CDM pipeline reaches 0.8673 ± 0.0230 AUROC under 5-fold cross-validation. This final chapter summarises our contributions, discusses the potential impact of this work, and offers closing reflections on the broader implications for machine learning.

8.1. Summary of Contributions

This research makes several interconnected contributions spanning theory, methodology, empirical evaluation, and practical deployment across two experimental tracks.

8.1.1. Addressing the Research Questions

We return to the six research questions posed in Chapter 1 and summarise the answers provided by our experimental investigation.

RQ1 (Effectiveness): Yes. On the within-CIFAR *airplane-vs-rest* split (a single ID class; multi-class generalisation is future work), the final selected $\lambda=0.02$ model reaches $99.03\% \pm 0.07\%$ AUROC over three seeds at $K=50$. The reproducible seed-42 $K=100$ evaluation reaches 98.98% AUROC and 90.50–96.97% across five reproducible external OOD datasets, evaluated against the full CIFAR-10 test reference pool used in the external protocol (Section 6.1), confirming reconstruction error as a robust distributional-fit signal within this scope.

RQ2 (Separation Loss): The separation loss mainly improves stability. Within-CIFAR airplane-vs-rest AUROC rises by +6.5 pp, from $92.52\% \pm 11.07\%$ at $\lambda = 0$ to $99.03\% \pm 0.07\%$ at $\lambda = 0.02$, both averaged over seeds 42, 123, and 456 (Section 6.4). The $\lambda = 0$

baseline is highly seed-sensitive (79.73%, 98.81%, 99.01%); a bootstrap 95% CI for the three-seed mean is [91.96%, 93.06%].

RQ3 (Monte Carlo Trials): Performance rises sharply to $K = 10$ (98.2% AUROC at $5\times$ speedup over $K = 50$) then saturates (Section 6.3). This design ablation used the seed-42 $\lambda = 0.01$ checkpoint for compute-budget reasons, while the final selected model uses $\lambda = 0.02$. $K = 10$ is recommended for most deployments; $K = 1$ (91.0%) serves as a fast fallback. Transferability of the K -AUROC curve to $\lambda = 0.02$ was not independently verified.

RQ4 (Scoring Strategy): Contrastive scoring is essential; in the seed-42 $\lambda = 0.01$ scoring ablation, ID-error-only scoring collapses on SVHN (20.2% AUROC). Difference scoring is retained as the default for its lower FPR95 and simpler interpretation (Section 6.3).

RQ5 (Practical Viability): Inference cost is higher than discriminative baselines but manageable: $K = 10$ requires ~ 16 min per 10K images on an NVIDIA GeForce RTX 2080 Ti. The accuracy–cost trade-off is favourable when the cost of a missed OOD sample exceeds the inference overhead.

RQ6 (Industrial Application): The public YOLO+CDM pipeline provides a reproducible industrial test case for conditional diffusion scoring, reaching 0.8673 ± 0.0230 AUROC at $\lambda = 0$ under 5-fold cross-validation. Unlike on CIFAR-10, the separation loss shows no statistically significant effect on this smaller dataset, and feature difficulty remains strongly heterogeneous (Section 6.6).

8.1.2. Main Contributions

Conceptual Contribution: We established a framework for using conditional diffusion models as generative classifiers, connecting diffusion models to generative classification theory. We demonstrated that class-conditional reconstruction error provides a principled signal for both OOD detection and quality classification.

Methodological Contribution: We developed diffusion-based detection and classification methods, including the multi-head conditioning mechanism for structured industrial data, differential noise prediction for binary quality classification, and systematic design guidelines for both tasks.

Empirical Contribution: Comprehensive evaluation across two tracks (CIFAR-10 OOD detection in Chapters 6–7, and public inkjet CDM quality classification under 5-fold cross-validation), producing reproducible results and clearly scoped ablation studies.

Practical and Implementation Contribution: Actionable deployment guidelines (separation loss for large-scale OOD; explicit public reproduction paths for industrial CDM evaluation) and modular public implementations built with PyTorch Lightning and Hydra. The CIFAR-10/OOD track is available at <https://github.com/ahmed-3m/DiffusionOOD> with weights at <https://huggingface.co/ahmed-3m/DiffusionOOD>;

the inkjet CDM track is available at <https://github.com/ahmed-3m/InkjetOOD> with weights at <https://huggingface.co/ahmed-3m/InkjetOOD>, and uses the public FTI_Zer0P dataset (Profactor, 2024).

8.2. Closing Remarks

This thesis demonstrates that conditional diffusion models, by learning rich representations of data manifolds, provide a versatile tool for both OOD detection and quality classification. Two overarching insights emerge: first, how a model is trained (separation loss, contrastive scoring) can matter as much as the architecture itself; second, successful transfer across domains depends on dataset scale, class separation, and evaluation protocol rather than on architectural reuse alone. These findings should be read as strong evidence within the two evaluated settings (a binary CIFAR-10 benchmark and the public FTI_Zer0P inkjet quality-control dataset), rather than as universal guarantees. Broader claims will require multi-class evaluation, adversarial stress tests, and replication on additional industrial datasets. For trustworthy deployment, the practical lesson is to optimise the training objective when learning distributional boundaries, then validate that objective again under the constraints of the target domain.

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A. Story Summary

A.1. Story Points

Q1. What is the central question? How can conditional diffusion models be effectively leveraged as generative classifiers for out-of-distribution detection, and what design choices, particularly regarding the training objective (separation loss) and inference strategy (Monte Carlo timestep sampling, contrastive scoring), most strongly influence detection performance?

Q2. Why is this question important? Deployed deep learning models encounter inputs outside their training distribution yet produce overconfident predictions, creating dangerous failure modes in safety-critical applications such as healthcare, autonomous driving, and manufacturing. Existing generative OOD detectors suffer from the likelihood paradox, where models assign higher likelihood to OOD data than to their own training data. Diffusion models offer reconstruction-based scoring that sidesteps this failure mode, but systematic investigation of their design space for OOD detection is lacking.

Q3. What evidence/data (variables) are needed to answer this question? We need OOD detection metrics (AUROC, FPR@95%TPR, AUPRC) across diverse near- and far-OOD benchmarks (CIFAR-100, Places365, FashionMNIST, SVHN, Textures) using a binary conditional diffusion model on CIFAR-10. Additionally, we require ablation results for the separation loss coefficient λ , the number of Monte Carlo trials K , and the scoring strategy (contrastive vs. ID-only). For industrial validation, we need public quality-classification AUROC, per-feature results, and separation-loss ablations from the YOLO+CDM inkjet pipeline under 5-fold cross-validation, all reproducible from Inkjet00D, public FTI_Zer0P data, and released CDM weights.

Q4. What methods are used to get this evidence/data? We train a binary conditional diffusion model (DDPM with UNet backbone) on CIFAR-10 with class-conditional noise prediction and a novel separation loss that maximises the distance between class-conditional noise predictions, indirectly widening reconstruction-error gaps. OOD scoring uses Monte Carlo timestep sampling with contrastive difference scoring. For the

industrial track, we build a public two-stage YOLO + CDM pipeline with multi-head conditioning and evaluate it using rigorous 5-fold cross-validation with image-level splitting.

Q5. What analyses must be applied to the data to answer the central question?

Systematic ablation studies isolate the effect of each design choice: a λ -sweep ($\{0.0, 0.001, 0.01, 0.02, 0.05, 0.1\}$) quantifies the separation loss contribution; a K -sweep ($\{1, 5, 10, 25, 50, 100\}$) characterises the accuracy–efficiency frontier; and a scoring strategy comparison (ID-only vs. difference vs. ratio) establishes the necessity of contrastive scoring. Cross-domain analysis compares the separation loss effect between CIFAR-10 and inkjet data to identify boundary conditions. Statistical testing (paired t -tests with Holm correction for $n \geq 5$ folds) validates industrial results.

Q6. What evidence/data (values for the variables) were obtained? On CIFAR-10, the binary CDM achieves 98.98% AUROC on the within-CIFAR split and 90.50–96.97% across five external OOD datasets. The separation loss improves the three-seed Within-CIFAR mean by +6.5 pp (from $92.52\% \pm 11.07\%$ at $\lambda=0$ to $99.03\% \pm 0.07\%$ at $\lambda=0.02$) while substantially reducing seed sensitivity. On inkjet data, the public YOLO+CDM pipeline reaches 0.8673 ± 0.0230 AUROC at $\lambda=0$ under 5-fold cross-validation, with no statistically significant gain from non-zero separation-loss settings. Per-feature analysis shows that distance and dot features are easiest, while edge roughness features remain more difficult and variable.

Q7. What were the results of the analyses? For CIFAR-10 OOD detection, the separation loss is the clearest performance and stability driver: binary conditioning alone can work well but is highly seed-sensitive, while $\lambda=0.02$ raises the three-seed mean to $99.03\% \pm 0.07\%$. Contrastive scoring is essential, since using ID error alone collapses to 20.2% AUROC on SVHN. $K=10$ Monte Carlo trials provide the best accuracy–efficiency trade-off (98.2% AUROC at $5\times$ speedup over $K=50$). On inkjet data, the separation loss shows no statistically significant effect (Holm-adjusted $p > 0.05$), revealing that its benefit requires sufficient dataset scale and inter-class visual difference.

Q8. How did the analyses answer the central question? Conditional diffusion models are effective generative classifiers for OOD detection when three design choices are made correctly in the evaluated CIFAR-10 setting: (1) the training objective benefits from a separation loss that widens class-conditional reconstruction-error gaps; (2) contrastive scoring comparing reconstruction under both conditions is essential; and (3) Monte Carlo timestep averaging with $K \geq 10$ is needed for reliable performance. The industrial application confirms that reconstruction error generalises as a quality signal, but also shows that the separation-loss gain does not transfer automatically to smaller

domain-specific datasets. On the inkjet industrial track, the public evidence supports YOLO+CDM performance, feature-level heterogeneity, and the negative separation-loss result, while broader method-ranking claims require complete public artefacts before being retained.

Q9. What does this answer tell us about the broader field? For tasks centred on distributional fit, such as OOD detection and quality classification, generative approaches modelling $p(x|y)$ through diffusion processes can provide strong, interpretable signals. The CIFAR-10 results suggest that objective-function engineering can matter as much as architecture when learning distributional boundaries. The negative separation-loss result on public inkjet data highlights that techniques validated on large-scale benchmarks do not automatically transfer to small domain-specific datasets, underscoring the importance of cross-domain validation.

B. Mathematical Derivations

This appendix provides mathematical derivations supporting the theoretical framework presented in Chapter 3. Standard DDPM derivations (forward process marginal, reverse posterior, ELBO decomposition, and the equivalence between noise prediction and denoising score matching) follow Ho et al. (2020) and Y. Song et al. (2021) directly and are not reproduced here. We focus on the results specific to our OOD detection framework.

B.1. Score-Based OOD Detection

For OOD detection, we consider how well the learnt score function $\nabla_x \log p_\theta(x)$ describes a test sample.

The reconstruction error can be interpreted as measuring the mismatch between the data and the learnt score function. Specifically, when we compute:

$$e(x) = \mathbb{E}_{t,\epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right] \quad (\text{B.1})$$

This measures how well the learnt score function can “denoise” the sample at various noise levels. OOD samples have scores that differ from the learnt distribution, leading to higher reconstruction error.

B.2. Reconstruction Error Analysis

B.2.1. Expected Reconstruction Error for In-Distribution Samples

For an in-distribution sample $x \sim p_{\text{data}}(x)$, the expected reconstruction error is:

$$\mathbb{E}_{x \sim p_{\text{data}}} [e(x)] = \mathbb{E}_{x,t,\epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right] \quad (\text{B.2})$$

$$\approx \mathbb{E}_{x,t,\epsilon} \left[\left\| \epsilon - \epsilon_\theta^*(x_t, t) \right\|^2 \right], \quad (\text{B.3})$$

where ϵ_θ^* denotes the best achievable denoiser for the learnt model class. This quantity is expected to be smaller for samples drawn from the training distribution than for samples far from it, but it is not generally zero: finite model capacity, stochastic optimisation, label noise, and the irreducible posterior uncertainty of the denoising task all leave residual error.

B.2.2. Reconstruction Error for OOD Samples

For an OOD sample $x' \notin p_{\text{data}}(x)$, the model attempts to project it onto the learnt manifold. The reconstruction error reflects the distance from the manifold:

$$e(x') = \mathbb{E}_{t,\epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x' + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|^2 \right] \quad (\text{B.4})$$

Under the low-dimensional support view, if in-distribution data concentrates near a structure $\mathcal{M} \subset \mathbb{R}^d$ and x' lies away from this region, the denoising model is expected to extrapolate less accurately than it does on typical ID samples. This motivates reconstruction error as an empirical OOD score, but it should be read as an intuition rather than a proof that error is strictly proportional to $d(x', \mathcal{M})$ for all OOD inputs.

B.2.3. Class-Conditional Reconstruction Error

For class-conditional models, the reconstruction error for class c is:

$$e_c(x) = \mathbb{E}_{t,\epsilon} \left[\left\| \epsilon - \epsilon_\theta \left(\sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c \right) \right\|^2 \right] \quad (\text{B.5})$$

For the binary CDM used in our experiments (class $c = 0$: in-distribution / airplane, class $c = 1$: out-of-distribution / non-airplane), the OOD score is computed via contrastive difference scoring:

$$s_{\text{OOD}}(x) = e_0(x) - e_1(x) \quad (\text{B.6})$$

This measures the relative reconstruction error advantage of the OOD-proxy condition over the ID condition. A large positive value indicates that the ID-conditioned model ($c = 0$) incurs higher reconstruction error than the OOD-conditioned model ($c = 1$), meaning the sample is better explained by the OOD-proxy class and is therefore out-of-distribution. Conversely, a negative value indicates the ID condition reconstructs the sample well, suggesting it is in-distribution. Using ID reconstruction error alone ($s = e_0(x)$) fails on datasets close to the training distribution, as absolute reconstruction magnitude is not discriminative without the contrastive reference.

B.3. Computational Complexity Analysis

B.3.1. Training Complexity

Training a diffusion model requires:

- Forward pass: $O(P)$ where P is the number of parameters
- Backward pass: $O(P)$
- Per training step: $O(B \cdot P)$ where B is batch size
- Total training: $O(N \cdot B \cdot P)$ where N is number of steps

For our binary conditional UNet with $P \approx 68.8\text{M}$ parameters, trained for $N \approx 200$ epochs on 50,000 training images (effective batch size 128 via gradient accumulation):

$$\text{FLOPs}_{\text{train}} \approx 50,000 \times 200 \times 2 \times 68.8\text{M} \approx 1.4 \times 10^{15} \text{ FLOPs} \quad (\text{B.7})$$

B.3.2. Inference Complexity

For OOD detection with K timestep trials and C conditioning classes:

- Single forward pass: $O(P)$
- Per sample: $O(K \cdot C \cdot P)$

For the binary CDM ($K = 50$, $C = 2$, $P \approx 68.8\text{M}$):

$$\text{FLOPs}_{\text{infer}} \approx 50 \times 2 \times 68.8\text{M} \approx 6.9 \times 10^9 \text{ FLOPs per sample} \quad (\text{B.8})$$

Compared to discriminative baseline (ResNet-18, $P = 11.2\text{M}$):

$$\text{FLOPs}_{\text{ResNet}} \approx 2 \times 11.2\text{M} = 2.24 \times 10^7 \text{ FLOPs per sample} \quad (\text{B.9})$$

The naïve parameter-count ratio yields $\sim 308\times$ overhead; however, this approximation assumes 2 FLOPs per parameter, which overestimates the effective computation for architectures with attention layers, skip connections, and conditional embeddings. On an NVIDIA GeForce RTX 2080 Ti, the measured wall-clock overhead is about $492\times$ for $K = 50$ and $68.6\times$ for $K = 10$ (Table D.2); $K = 1$ was not re-timed in the final consistency runtime check.

C. Experimental Configuration

This appendix provides detailed hyperparameters, hardware specifications, and configuration examples for reproducing our public experiments. The CIFAR-10/OOD implementation is publicly available at <https://github.com/ahmed-3m/DiffusionOOD> (commit 1b4ce82), with model weights at <https://huggingface.co/ahmed-3m/DiffusionOOD>. The inkjet CDM implementation and experiment documentation are publicly available at <https://github.com/ahmed-3m/InkjetOOD> (Mohammed, 2026b) (commit 7c2451c), with model weights at <https://huggingface.co/ahmed-3m/InkjetOOD>; the accompanying PROFACITOR FTI_Zer0P dataset is available on Zenodo under CC BY 4.0 (Profactor, 2024).

C.1. Model Architecture Hyperparameters

Table C.1 specifies architectural hyperparameters for the conditional UNet model.

Table C.1.: Architectural hyperparameters for conditional UNet model.

Parameter	Value	Description
Input channels	3	RGB images
Block channels	[128, 256, 256, 512]	Channel multipliers per block
Attention resolutions	[16, 8]	Resolutions with self-attention
Class embedding dim	512	Class embedding dimension
Timestep embedding dim	512	Timestep embedding dimension
Residual blocks per layer	2	ResNet blocks
Total parameters	~68.8M	Trainable parameters (binary CDM)
Model size (FP32)	~275 MB	Disk footprint (68.8M $\times$ 4 bytes)

Table C.2 lists CIFAR-10/OOD training hyperparameters. Table C.3 specifies CIFAR-10/OOD inference parameters. Table C.4 lists the retained inkjet CDM hyperparameters.

Table C.2.: Training hyperparameters for CIFAR-10/OOD experiments.

Parameter	Value	Description
Optimiser	AdamW	Optimiser algorithm
Learning rate	10^{-4}	Base learning rate
LR schedule	Cosine + 5-ep. warm-up	Learning rate schedule
Weight decay	$\lambda_{wd} = 0.01$	L_2 regularisation
Gradient clip	1.0	Max gradient norm
Batch size	128	Training batch size
Max epochs	200 (patience = 30)	Typical conv. ep. 15–25
Diffusion timesteps T	1000	Number of noise levels
Noise schedule	Cosine	Variance schedule
EMA decay	0.9999	Exponential moving average

Table C.3.: Hyperparameters for CIFAR-10/OOD inference.

Parameter	Value	Description
Timestep trials K	50	Samples for OOD scoring
Timestep sampling	Uniform	Sample strategy
Batch size	256	Inference batch size
Use EMA weights	True	Use moving average model

Table C.4.: Retained inkjet CDM hyperparameters.

Parameter	Value	Description
Model	NoisePredictorV3	Custom UNet with multi-head conditioning
Base channels	64	Approximately 9.3M parameters
Input resolution	128×128	YOLO-cropped feature patches
Conditioning heads	4	Template, feature type, quality label, and bounding box
Optimiser	AdamW	Optimiser algorithm
Learning rate	2×10^{-4}	Inkjet CDM training rate
Weight decay	10^{-4}	Inkjet CDM regularisation
Batch size	128 / 64	Baseline $\lambda = 0$ / separation-loss $\lambda > 0$ runs
Epochs	100 per fold	Retained 5-fold CV configuration
Diffusion timesteps T	1000	Number of noise levels
Noise schedule	Cosine	Retained public implementation default
Timestep trials K	100	Final reported 5-fold CV ablation; implementation default is 50
Folds / seed	5 / 42	Stratified image-level cross-validation

C.2. Computational Resources

C.2.1. Hardware and Software Environment

Table C.5.: Hardware specifications.

Component	Specification
GPUs	NVIDIA V100 (16 GB), GV100 (32 GB), P40 (24 GB)
GPU config	Single GPU per run, mixed precision (AMP)
CPU	Intel Xeon Gold 6154 / 6136 (36 cores, 3.00 GHz)
RAM	378–384 GB DDR4

Table C.6.: Software environment and library versions.

Software	Version
Python	3.10
PyTorch	2.1.x
PyTorch Lightning	2.1.x
Diffusers	0.25.x
Hydra	1.3.x
CUDA	12.1
cuDNN	8.9.x

C.2.2. Training Time and Memory

Experiments were run on whichever GPU was available: NVIDIA V100 (16 GB), GV100 (32 GB), or P40 (24 GB), always one GPU per run, batch size 64 with gradient accumulation $\times 2$ (effective batch 128). Approximate wall-clock time per seed: ~ 10 – 16 h on GV100, ~ 18 – 28 h on V100 or P40. Typical convergence occurs at epoch 15–25 of the 200-epoch maximum.

Memory usage: Model FP32 (~ 0.28 GB) + AdamW states (~ 0.55 GB) + activations (~ 6 – 10 GB at batch 64) ≈ 7 – 11 GB in full precision. With mixed-precision training (AMP), peak memory drops to ≈ 5 – 7 GB, comfortably within the V100’s 16 GB, the tightest GPU used.

C.3. Configuration Examples

C.3.1. Training Configuration (Hydra)

```
# config/experiment/exp_baseline.yaml
defaults:
  - override /model: conditional_diffusion
  - override /data: cifar10
  - override /trainer: base_trainer

model:
  learning_rate: 1e-4
  num_timesteps: 1000
  ema_decay: 0.9999

data:
  batch_size: 128
  num_workers: 8

trainer:
  max_epochs: 200
  precision: 16
  accelerator: gpu
  devices: 1

seed: 42
```

C.4. Reproducibility

Table C.7.: Public artefacts required to reproduce the thesis experiments.

Component	Public source	Reproduction role
CIFAR-10/OOD scripts	https://github.com/ahmed-3m/DiffusionOOD	Training, scoring, ablation, and figure-generation scripts for the public CIFAR-10 OOD experiments.
CIFAR-10/OOD weights	https://huggingface.co/ahmed-3m/DiffusionOOD	Pre-trained checkpoints for retained seed/ λ configurations; enables direct evaluation without retraining.
Inkjet CDM scripts	https://github.com/ahmed-3m/InkjetOOD (Mohammed, 2026b), commit 7c2451c	Training, evaluation, 5-fold CV, separation-loss ablation, and figure-generation scripts for the retained inkjet CDM experiments.
Inkjet CDM weights	https://huggingface.co/ahmed-3m/InkjetOOD	Pre-trained inkjet CDM checkpoint; enables direct evaluation on the FTI_Zer0P dataset without retraining.
Inkjet dataset	PROFACTOR FTI_Zer0P on Zenodo (Profactor, 2024)	Public image data and metadata used by the inkjet reproduction scripts; released under CC BY 4.0.
Figure regeneration	Inkjet00D/scripts/ and Diffusion00D plotting scripts	Regenerates retained result figures from public raw scores or documented evaluation outputs.

Primary evaluation and ablation experiments use random seed 42. Robustness checks for the separation loss ablation ($\lambda \in \{0.01, 0.02\}$) are performed across three seeds (42, 123, 456) as described in Section 6.1. The seed is initialised with:

```
import random, numpy as np, torch

def set_seed(seed=42):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    torch.cuda.manual_seed_all(seed)
    torch.backends.cudnn.deterministic = True
    torch.backends.cudnn.benchmark = False
```

Complete experiment metadata is logged to Weights & Biases including: Git commit hash, all hyperparameters, training/validation metrics per epoch, hardware utilisation, and checkpoint paths.

Dataset versions: CIFAR-10 (torchvision 0.15.2), CIFAR-100 (torchvision 0.15.2), SVHN (format 2, cropped), Textures (DTD all 47 categories), FashionMNIST (torchvision, re-

sized to 32×32 , greyscale to RGB), and Places365 (10,000 scene images resized to 32×32).

All code is version controlled via Git with commit hashes logged for every experiment. Configuration files (including all command-line overrides) are saved with results for exact reproduction.

C.4.1. Audit-Policy Summary

Table C.8 consolidates which evaluation protocol backs each retained claim family, so that the seed coverage, cross-validation protocol, checkpoint λ , and Monte Carlo trial budget K are visible at a glance. This table is the canonical reference for the audit boundary referred to throughout Chapters 6 and 7.

Table C.8.: Audit-policy summary. For each retained claim family, the seeds evaluated, cross-validation protocol, separation-loss weight λ of the checkpoint used, and Monte Carlo trial budget K . “Single test split” refers to the standard CIFAR-10 train/test split with no resampling; the inkjet track uses 5-fold image-level stratified cross-validation over 573 source-image groups.

Claim family	Source	Seeds	Protocol	λ	K
CIFAR within-CIFAR headline	§6.4	42, 123, 456	single split	0.02	50
CIFAR external OOD (5 sets)	§6.1	42	single split	0.02	100
K -ablation	§6.3, App. D	42	single split	0.01	varies
Timestep-sampling ablation	§6.3	42	single split	0.01	50
Scoring-strategy ablation	§6.3, App. D	42	single split	0.01	50
Inkjet CDM (λ -sweep)	§6.6	42 (single)	5-fold strat.	0, 0.01, 0.02, 0.05	50

Two design asymmetries are deliberate and documented above: (i) the K -, timestep-, and scoring-strategy ablations use the seed-42 $\lambda=0.01$ checkpoint rather than the final $\lambda=0.02$ model for compute-budget reasons, and (ii) the external OOD evaluation is single-seed (seed 42) but uses $K=100$ to reduce Monte Carlo variance in the reported means. Both are surfaced verbatim in Chapter 6 where the affected numbers first appear.

D. Additional Results

This appendix provides supplementary ablation results that support Chapter 6. All reported values are tied to retained public artefacts. The complete OOD detection results (including FPR@95%TPR) are presented in the main text (Table 6.1).

D.1. Separation Loss Ablation: Detailed Results

Table D.1 shows the effect of the separation loss coefficient λ on within-CIFAR OOD detection performance. All results use $K=50$ MC trials and difference scoring. The $\lambda \in \{0.00, 0.01, 0.02\}$ entries are mean $\pm$ std over three seeds.

Table D.1.: Separation loss ablation: AUROC (%) on within-CIFAR split as a function of λ . All conditions use $K=50$ and difference scoring. The $\lambda \in \{0.00, 0.01, 0.02\}$ rows report three seeds (42/123/456).

λ	AUROC (%) $\uparrow$	Δ vs. $\lambda=0$
0.00 (no separation loss)	92.52 ± 11.07	—
0.01	98.82 ± 0.06	+6.30
0.02	99.03 ± 0.07	+6.51

The results show that the main effect of a non-zero λ is reduced seed sensitivity. The peak is achieved at $\lambda=0.02$, which is the best/primary reported setting. The $\lambda=0.01$ checkpoint is retained for the K , timestep-sampling, and scoring-strategy ablations because those runs were completed before the final three-seed sweep and were not rerun at $\lambda=0.02$. Larger λ values risk over-constraining the model, which we leave to future work.

D.2. Monte Carlo Trials Ablation: Detailed Results

Table D.2 characterises the accuracy–efficiency frontier as a function of the number of MC timestep trials K .

Table D.2.: K ablation: AUROC (%) and relative inference cost on within-CIFAR split using the seed-42 $\lambda=0.01$ checkpoint. Reference cost is the discriminative baseline (ResNet-18 forward pass) measured on an NVIDIA GeForce RTX 2080 Ti. The measured wall-clock timings are 972.9 s for $K=10$ and 4861.1 s for $K=50$ per 10K images.

K	AUROC (%) $\uparrow$	Inference Cost	Notes
1	91.0	not re-timed in final consistency pass	Low latency
10	98.2	$68.6\times$ discriminative	Recommended trade-off
50	98.64	$492.4\times$ discriminative	Default (within-CIFAR)

The performance gain from $K=1$ to $K=10$ is substantial (+7.2 pp), while the gain from $K=10$ to $K=50$ is marginal (+0.44 pp). This suggests that $K=10$ is the practical sweet spot for most deployment scenarios. For the external OOD evaluation, $K=100$ was used to maximise statistical stability of the reported means; $K=50$ produces similar AUROC values on those datasets.

D.3. Scoring Strategy: Difference vs. ID-Only

Table D.3 compares contrastive difference scoring ($S = e_0(x) - e_1(x)$, where $c = 0$ is ID and $c = 1$ is non-ID) against single-sided ID-only scoring ($S = e_0(x)$) on selected OOD datasets.

Table D.3.: Scoring strategy comparison using the seed-42 $\lambda=0.01$ checkpoint at $K=50$: AUROC (%) for contrastive difference, ratio, and ID-only scoring.

Dataset	Difference	Ratio	ID-Only	Best – ID-Only
Within-CIFAR	98.69	98.62	78.30	+20.39
SVHN	94.13	96.06	20.2	+75.86

All values come from the retained seed-42 $\lambda=0.01$ scoring ablation JSON at $K=50$. The main text reports difference scoring because it is the default reproducible scorer.

The catastrophic failure of ID-only scoring on SVHN (20.2% AUROC, below random chance on a balanced evaluation) occurs because SVHN images have lower absolute reconstruction error under the ID model than many within-CIFAR OOD images, making the absolute magnitude uninformative. Contrastive scoring resolves this by measuring the relative advantage of the ID model, providing a signal that is robust to absolute reconstruction magnitude differences. Ratio scoring is competitive here, but the thesis retains difference scoring as the primary reproducible scorer for consistency across tables and raw-score evaluation.